UNIVERSIDADE DE SÃO PAULO
Instituto de Ciências Matemáticas e de Computação

Model of theses and dissertations in LaTeX of the ICMC

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Qualificação de Mestrado do Programa Interinstitucional de Pós-Graduação em Estatística (PIPGEs)

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Gabriel de Jesus Pereira

Model of theses and dissertations in LaTeX of the ICMC

Monograph submitted to the Institute of Mathematics and Computer Science – ICMC-USP and to the Department of Statistics – DEs-UFSCar – as part of the qualifying exam requisites of the Statistics Interagency Graduate Program, for the degree of Master in Statistics.

Concentration Area: Statistics

Advisor: Profa. Dra. Vera Lucia Damasceno Tomazella

USP – São Carlos
August 2021

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A634m Antonelli, Humberto Lidio
Modelo de teses e dissertações em LaTeX do ICMC / Humberto Lidio Antonelli; orientador Renata Pontin de Mattos Fortes. -- São Carlos, 2018.
81 p.

Tese (Doutorado - Programa de Pós-Graduação em Ciências de Computação e Matemática Computacional) -- Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, 2018.

1. Modelo. 2. Monografia de qualificação. 3. Dissertação. 4. Tese. 5. Latex. I. Fortes, Renata Pontin de Mattos, orient. II. Título.

Bibliotecários responsáveis pela estrutura de catalogação da publicação de acordo com a AACR2:
Gláucia Maria Saia Cristianini - CRB - 8/4938
Juliana de Souza Moraes - CRB - 8/6176

Gabriel de Jesus Pereira

Título - A Definir

Monografia apresentada ao Instituto de Ciências Matemáticas e de Computação – ICMC-USP e ao Departamento de Estatística – DEs-UFSCar, para o Exame de Qualificação, como parte dos requisitos para obtenção do título de Mestre em Estatística – Programa Interinstitucional de Pós-Graduação em Estatística.

Área de Concentração: Estatística

Orientadora: Profa. Dra. Vera Lucia Damasceno Tomazella

USP – São Carlos
Agosto de 2021

DEDICATÓRIA!!!!

ACKNOWLEDGEMENTS

AGRADECIMENTOS!!!!

“EPÍGRAFE!!!!”

RESUMO

GABRIEL, J. P. **Título - A Definir**. 2021. 73 p. Monografia (Mestrado em Estatística – Programa Interinstitucional de Pós-Graduação em Estatística) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2021.

Este artigo introduz o GG-KM, uma nova abordagem semiparamétrica para modelagem de taxa de cura no modelo de tempo de promoção. Abordamos as limitações dos preditores lineares tradicionais ao empregar Máquinas de Kernel para modelar o componente de cura dentro de um Reproducing Kernel Hilbert Space (RKHS) (MOHRI; ROSTAMIZADEH; TALWALKAR, 2018). Para fornecer máxima flexibilidade à função de risco dos indivíduos suscetíveis, incorporamos a distribuição Gama Generalizada (STACY, 1962), que engloba diversos modelos clássicos de sobrevivência. Derivamos uma estrutura de log-verossimilhança penalizada e fornecemos gradientes analíticos explícitos para a otimização conjunta dos pesos do kernel, hiperparâmetros e parâmetros da distribuição. O desempenho do modelo é validado por meio do Integrated Brier Score (IBS) com ajuste por IPCW (PRINCE *et al.*, 2025).

Palavras-chave: Análise de Sobrevivência, Censura, Reproducing Kernel Hilbert Space, Gama Generalizada, Representer Theorem.

ABSTRACT

GABRIEL, J. P. **Model of theses and dissertations in LaTeX of the ICMC**. 2021. 73 p. Monografia (Mestrado em Estatística – Programa Interinstitucional de Pós-Graduação em Estatística) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2021.

This paper introduces GG-KM, a novel semiparametric approach to promotion time cure rate modeling. We address the limitations of traditional linear predictors by employing Kernel Machines to model the cure component within a Reproducing Kernel Hilbert Space (RKHS) (MOHRI; ROSTAMIZADEH; TALWALKAR, 2018). To provide maximum flexibility for the hazard of susceptible individuals, we incorporate the Generalized Gamma (GG) distribution (STACY, 1962), which encompasses several standard survival models. We derive a penalized log-likelihood framework and provide explicit analytical gradients for the joint optimization of kernel weights, hyperparameters, and distribution parameters. Model performance is validated through the Integrated Brier Score (IBS) with IPCW adjustment (PRINCE *et al.*, 2025).

Keywords: Survival Analysis, Censoring, Reproducing Kernel Hilbert Space, Generalized Gamma, Representer Theorem.

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FOUNDATIONS OF SURVIVAL ANALYSIS

2.1 Introduction

Knowing whether an event of interest will occur is crucial. However, in many situations, knowing when it will occur can be even more informative. In this context, a statistical methodology known as survival analysis arises. The elapsed time until the occurrence of an event of interest is referred to as the survival time (AALEN; BORGAN; GJESSING, 2008). For example, in a study concerned with loan default among credit union members, the survival time may be defined as the time until a member defaults on a loan. Survival analysis is therefore concerned with modeling the distribution of this time and investigating factors that may influence its occurrence. Thus, the objective of this chapter is to present the fundamental concepts underlying survival analysis.

2.2 Censoring

The main characteristic that distinguishes survival analysis from other areas of statistics is the presence of censored data. For example, consider a study in which credit union members are followed over time to investigate the occurrence of loan default. In this context, two situations may happen. Some members may default during the follow-up period, whereas others may not experience the event before the end of the study. In the latter case, censoring occurs because the survival time of these individuals is not fully observed; it is only known that the event will occur after a certain point in time.

Censoring can occur in three different forms. The example described above can be regarded as right censoring. Right censoring occurs when the event takes place after a certain time, but its exact occurrence time is unknown. Another type is left censoring, which occurs when the event has taken place before a certain time. In the context of loan default, an example of left censoring would be a member who was already in default at the beginning of the study, while the exact time at which the default occurred is unknown. Finally, interval censoring occurs when the event takes place within a given time interval, but its exact occurrence time is unknown. Given the loan default example, interval censoring could arise if members were evaluated periodically and default was detected only during these assessments, without knowledge of the exact time at which it occurred. Figure 1 illustrates each type of censoring.

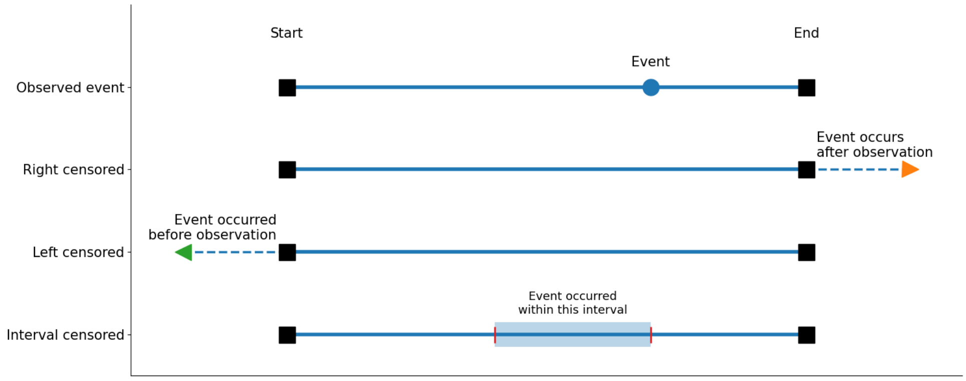

Figure 1 – Illustration of observed events and censoring types in survival analysis: observed event, right censoring, left censoring, and interval censoring.

Formally, let T denote the survival time and let l and u denote the lower and upper bounds of the censoring interval, respectively. Under interval censoring, it is only known that $l < T \leq u$, where both l and u are finite. Notice that left and right censoring can be viewed as special cases of interval censoring. Hence, left censoring corresponds to the interval $(-\infty, u]$, whereas right censoring corresponds to the interval (l, ∞).

The methods presented in this work consider only right censoring, since it is the most common type encountered in practice. Thus, to represent survival data, let T^* be a random variable representing the time to failure, i.e., the time until the occurrence of the event of interest, and let U be a random variable representing the time to a censoring event. The observed time is $T = \min(T^*, U)$ and a censoring indicator $\delta = I[T^* \leq U]$. That is, δ takes the value 0 or 1 according to whether T is an observed failure time or a censored time (MOORE *et al.*, 2016). Hence, throughout this work, survival data will be represented by the triplet $\{(\mathbf{x}_i, t_i, \delta_i)\}_{i=1}^{n}$, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the vector of covariates, t_i the observed time, and δ_i an indicator variable associated with the i-th individual. The sample size is denoted by n, and p denotes the number of covariates.

Therefore, survival data are generally composed of both complete and incomplete observations (AALEN; BORGAN; GJESSING, 2008). As a result, survival analysis methods seek to incorporate censored observations into their methodology. Given the definition of censoring presented in this section, the next section introduces the definitions of the survival function and hazard rate, as well as some relationships between these functions.

2.3 Survival and hazard rate functions

Survival analysis is concerned with the study of the time until the occurrence of an event of interest. A survival distribution can be characterized through several equivalent functions, among which the survival function and the hazard rate function are fundamental (MOORE *et al.*, 2016). The survival function describes the probability that the event has not occurred by a given time t. Formally, let T denote the nonnegative random variable representing the survival time. The survival function is defined as

$$S(t) = P(T > t), \qquad t \geq 0. \tag{2.1}$$

Thus, $S(t)$ represents the probability that an individual survives beyond time t, or, equivalently, that the event of interest has not occurred by that time. The survival function is a non-increasing function satisfying $S(0) = 1$ and $\lim_{t\to\infty} S(t) \geq 0$.

In the usual setting, the survival function decreases as t increases, reflecting the accumulation of events over time. However, it need not converge to zero. In particular, if $\lim_{t\to\infty} S(t) = 0$, the corresponding distribution is said to be proper. Conversely, when $\lim_{t\to\infty} S(t) > 0$, the distribution is said to be improper. The latter case is of particular relevance in cure-rate models, where the positive limiting value of the survival function can be interpreted as the probability of an individual belonging to a cured fraction. This distinction will be discussed in greater detail in the next section.

The hazard rate function, denoted by $h(t)$, characterizes the instantaneous risk of experiencing the event of interest, conditional on survival up to time t. More precisely, assuming that T is an absolutely continuous random variable with probability density function $f(t)$, the hazard rate at time t is defined as

$$h(t) = \lim_{\Delta t \to 0} \frac{1}{\Delta t} P\left(t \leq T < t + \Delta t \mid T \geq t\right). \tag{2.2}$$

Thus, for a sufficiently small time interval Δt, the quantity

$$h(t)\Delta t$$

provides a first-order approximation to the conditional probability of experiencing the event during the interval $[t, t + \Delta t)$, given that the individual has survived up to time t. It is important to note that the hazard rate itself is not a probability and, consequently, is not restricted to the interval $[0, 1]$.

Under the assumption of absolute continuity, the hazard rate can be expressed in terms of the probability density function and the survival function as

$$h(t) = \frac{f(t)}{S(t)}. \tag{2.3}$$

Conversely, the survival function can be recovered from the hazard rate through the cumulative hazard function, defined as

$$H(t) = \int_0^t h(u)\,du. \tag{2.4}$$

Consequently,

$$S(t) = \exp\{-H(t)\} = \exp\left\{-\int_0^t h(u)\,du\right\}. \tag{2.5}$$

Equation (2.5) establishes the fundamental relationship between the hazard rate and the survival function. In particular, the survival function is determined by the cumulative hazard over time. Consequently, different hazard profiles may lead to substantially different survival patterns, even when they are defined on the same time scale.

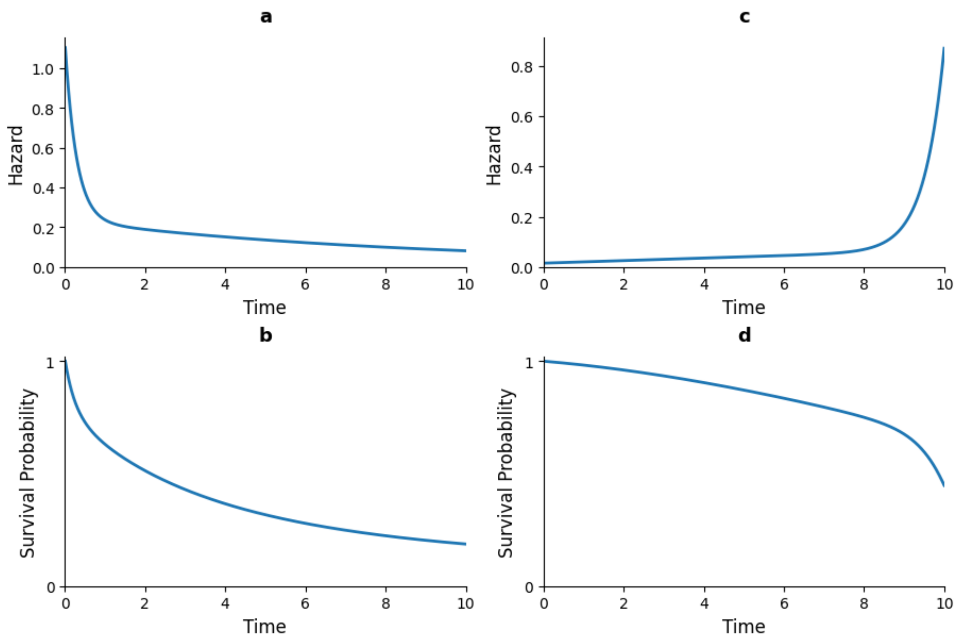

Figure 2 – Illustration of observed events and censoring types in survival analysis: observed event, right censoring, left censoring, and interval censoring.

Figure 2 illustrates this relationship. In panels (a) and (b), the hazard rate is relatively high at early times, resulting in a more pronounced decline in the corresponding survival function. Such a pattern may arise, for example, in populations characterized by a high risk of death shortly after entering the study. In contrast, panels (c) and (d) illustrate a hazard with a lower initial risk, leading to a slower decline in the corresponding survival function. These examples emphasize that the shape of the survival function is directly determined by the evolution of the hazard rate over time.

Although the survival and hazard rate functions provide two fundamental ways of characterizing a survival distribution, other distributional functions are also useful, particularly when deriving likelihoods and establishing relationships among different representations of the survival distribution. Among these, the cumulative distribution function and the probability density function are of particular importance.

The cumulative distribution function (CDF) of the survival time T is defined by

$$F(t) = P(T \leq t), \tag{2.6}$$

and, for a nonnegative random variable, is related to the survival function through

$$F(t) = 1 - S(t). \tag{2.7}$$

When T is absolutely continuous, its probability density function (PDF) is given by the derivative of the CDF,

$$f(t) = \frac{d}{dt}F(t) = -\frac{d}{dt}S(t). \tag{2.8}$$

Consequently, the principal functions associated with an absolutely continuous survival distribution are connected through

$$f(t) = h(t)S(t),$$

which follows directly from the definition of the hazard rate.

These relationships provide equivalent representations of the same underlying survival distribution. In particular, knowledge of any one of the functions, under the appropriate regularity conditions, allows the others to be obtained. The relationships among these functions will be used throughout the subsequent development of the survival and cure-rate models considered in this work.

2.3.1 Likelihood for censored survival times

In the absence of censoring, suppose that the survival times $T_1^*, \ldots, T_n^*$ are independent and identically distributed with probability density function $f(t;\boldsymbol{\theta})$ and survival function $S(t;\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ denotes the vector of model parameters. The likelihood function based on the complete survival times is then given by

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} f(t_i^*;\boldsymbol{\theta}). \tag{2.9}$$

In practice, however, survival times are frequently subject to right censoring. As defined previously, let T^* denote the time to failure and U the censoring time. The observed time and censoring indicator are given by $T = \min(T^*, U)$ and $\delta = I[T^* \leq U]$, respectively. Thus, $\delta = 1$ indicates that the event of interest was observed, whereas $\delta = 0$ indicates that the observation was right-censored.

Accordingly, the observed survival data considered throughout this work are represented by the triplet $\{(\boldsymbol{x}_i, t_i, \delta_i)\}_{i=1}^n$, where $\boldsymbol{x}_i \in \mathbb{R}^p$ denotes the vector of covariates for the ith individual, t_i is the observed time, and δ_i is the corresponding event indicator.

For an individual with $\delta_i = 1$, the event is observed at time t_i, and the likelihood contribution is given by the density $f(t_i;\boldsymbol{\theta})$. In contrast, when $\delta_i = 0$, the event has not occurred by time

t_i, and the available information is that $T_i^* > t_i$. Consequently, the likelihood contribution is given by the survival probability $S(t_i;\boldsymbol{\theta})$. These two cases can be combined into the single expression

$$L_i(\boldsymbol{\theta}) = f(t_i;\boldsymbol{\theta})^{\delta_i} S(t_i;\boldsymbol{\theta})^{1-\delta_i}. \tag{2.10}$$

Assuming independence among individuals, the likelihood function based on the observed survival data is therefore

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} f(t_i;\boldsymbol{\theta})^{\delta_i} S(t_i;\boldsymbol{\theta})^{1-\delta_i}. \tag{2.11}$$

Using the relationship $f(t;\boldsymbol{\theta}) = h(t;\boldsymbol{\theta})S(t;\boldsymbol{\theta})$, the likelihood can equivalently be expressed in terms of the hazard rate as

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} h(t_i;\boldsymbol{\theta})^{\delta_i} S(t_i;\boldsymbol{\theta}). \tag{2.12}$$

This formulation is fundamental in survival analysis because it accounts simultaneously for observed failures and right-censored observations. It will also provide the basis for the likelihood formulations considered later for the long-term survival and cure-rate models developed in this work.

In conventional survival models, it is typically assumed that every individual is eventually susceptible to the event, implying that $\lim_{t\to\infty} S(t) = 0$. However, this assumption may be inappropriate in settings where a subset of individuals is considered effectively immune to the event of interest. In such cases, the survival function may converge to a positive limiting value, $\lim_{t\to\infty} S(t) = \pi > 0$, where π represents the asymptotic probability of remaining event-free. This behavior is characteristic of cure-rate models and provides a natural framework for explicitly modeling a cured fraction within the population. The distinction between these two settings is central to the development of cure-rate models and will be considered in the following section.

2.4 Long-term survival models

In some survival studies, a non-negligible proportion of individuals may never experience the event of interest, even when followed for an arbitrarily long period of time. Such individuals are commonly referred to as cured or immune with respect to the event under consideration (MALLER; ZHOU, 1996). The presence of such individuals gives rise to the class of models commonly referred to as long-term survival models or cure-rate models.

This setting differs fundamentally from that of conventional survival models, which generally assume that every individual is susceptible to the event of interest and that the event will eventually occur with probability one. Under this assumption, the survival function satisfies

$$\lim_{t\to\infty} S(t) = 0. \tag{2.13}$$

In contrast, when a proportion of the population is not susceptible to the event, the population survival function may converge to a positive value,

$$\lim_{t\to\infty} S(t) = \pi, \qquad 0 < \pi < 1, \tag{2.14}$$

where π represents the long-term survival probability, commonly interpreted as the population cure fraction.

The formulation of long-term survival models considered in this work follows the extension proposed by Rodrigues *et al.* (2009). In this extension, the occurrence of the event of interest is represented through a collection of competing causes. Let N be a random variable denoting the number of competing causes related to the occurrence of the event of interest, with probability distribution

$$p_n = P(N = n), \qquad n = 0, 1, 2, \ldots. \tag{2.15}$$

Conditional on $N = n$, let $Z_1, \ldots, Z_n$ denote the latent times to the event associated with these causes. Assume that these random variables are independent and identically distributed, with proper survival function $S(t)$ that does not depend on N. In order to include individuals who are not susceptible to the event, the event time is defined as the minimum among the latent times,

$$T = \min\{Z_1, \ldots, Z_N\}, \tag{2.16}$$

with $P(Z_0 = \infty) = 1$, which leads to a proportion p_0 of the population not susceptible to the event occurrence.

Given the previous definitions, let $\{a_n\}$ be a real-valued sequence and $s \in [0, 1]$. The generating function (FELLER *et al.*, 1971), denoted by $A(s)$, of the sequence $\{a_n\}$ is given by

$$A(s) = a_0 + a_1 s + a_2 s^2 + \ldots. \tag{2.17}$$

The survival function of the random variable T is then obtained as

$$\begin{aligned} S_p(t) &= P(N=0) + P(Z_1 > t, Z_2 > t, \ldots, Z_N > t, N \geq 1) \\ &= P(N=0) + \sum_{n=1}^{\infty} P(N=n) P(Z_1 > t, Z_2 > t, \ldots, Z_n > t) \\ &= p_0 + \sum_{n=1}^{\infty} p_n \{S(t)\}^n \\ &= A(S(t)), \end{aligned} \tag{2.18}$$

where $A(\cdot)$ is the generating function associated with the probability mass function $\{p_n\}$, as defined in Equation (2.17), and $S(t)$ is a proper survival function. Since $\lim_{t\to 0} S(t) = 1$ and $\lim_{t\to\infty} S(t) = 0$, it follows that $\lim_{t\to\infty} S_p(t) = A(0) = P(N = 0) = p_0$.

Thus, when $p_0 > 0$, the long-term survival function has a positive limiting value, which can be interpreted as the proportion of individuals in the population who are not susceptible to the event of interest, i.e., the cure fraction. Consequently, $S_p(t)$ is an improper survival function. It is a non-increasing function and satisfies $S_p(0) = 1$.

The probability density function and the hazard rate function associated with the long-term survival function are given, respectively, by

$$f_p(t) = f(t)\frac{dA(s)}{ds}|_{s=S(t)}, \tag{2.19}$$

$$h_p(t) = \frac{f_p(t)}{S_p(t)} = f(t)\frac{\frac{dA(s)}{ds}|_{s=S(t)}}{S_p(t)}. \tag{2.20}$$

Table 1 – Long-term survival, density, hazard, and cure fraction for the different distributions considered for the latent number of competing causes N.

Distribution of N	$S_p(t)$	$f_p(t)$	$h_p(t)$	p_0
Bernoulli(θ)	$1-\theta+\theta S(t)$	$\theta f(t)$	$\frac{\theta f(t)}{1-\theta+\theta S(t)}$	$1-\theta$
Binomial $\left(K, \theta^* = \frac{\theta}{1+\theta}\right)$	$[1-\theta^*+\theta^* S(t)]^K$	$K\theta^* f(t)\,[1-\theta^*+\theta^* S(t)]^{K-1}$	$\frac{K\theta^* f(t)}{1-\theta^*+\theta^* S(t)}$	$(1-\theta^*)^K$
Negative Binomial(α,θ)	$\{1+\alpha\theta\,[1-S(t)]\}^{-1/\alpha}$	$\theta f(t)\{1+\alpha\theta\,[1-S(t)]\}^{-1/\alpha-1}$	$\frac{\theta f(t)}{1+\alpha\theta\,[1-S(t)]}$	$(1+\alpha\theta)^{-1/\alpha}$
Poisson(θ)	$\exp\{-\theta[1-S(t)]\}$	$\theta f(t)\exp\{-\theta[1-S(t)]\}$	$\theta f(t)$	$e^{-\theta}$

The methods developed in this work consider different distributions for the latent variable N, representing the number of competing causes. Specifically, the Bernoulli, Binomial, Negative Binomial, and Poisson distributions are considered. The corresponding results are summarized in Table 1, and their derivations are given below, beginning with the Bernoulli distribution.

2.4.1 Bernoulli distribution

Let N be a random variable representing the number of latent competing causes associated with the occurrence of a given event, following a Bernoulli distribution with parameter $\theta \in (0,1)$, i.e., $N \sim$ Bernoulli(θ). Its probability mass function is given by

$$P(N=n) = \theta^n(1-\theta)^{1-n}, \qquad n = 0, 1. \tag{2.21}$$

The generating function of N is given by

$$A(s) = 1-\theta+\theta s,$$

and, therefore, the long-term survival function is

$$S_p(t) = A(S(t)) = 1-\theta+\theta S(t). \tag{2.22}$$

This corresponds to the mixture cure model introduced by (BERKSON; GAGE, 1952), one of the most widely used models for long-term survival data. The model is referred to as a mixture cure model because the population survival function can be expressed as a mixture of a cure component and a proper survival function corresponding to the susceptible population. In this case, the cure fraction is given by

$$p_0 = \lim_{t\to\infty} S_p(t) = 1-\theta. \tag{2.23}$$

The probability density and hazard rate functions are given, respectively, by

$$f_p(t) = \theta f(t) \tag{2.24}$$

and

$$h_p(t) = \frac{\theta f(t)}{1-\theta+\theta S(t)}. \tag{2.25}$$

2.4.2 Binomial distribution

The Binomial distribution provides a natural extension of the Bernoulli specification by allowing the number of latent competing causes to take values from 0 to K. Let N be a random variable representing the number of latent competing causes associated with the occurrence of the event of interest, following a Binomial distribution with parameters K and $\theta^* \in (0,1)$, i.e.,

$$N \sim \text{Binomial}\left(K, \theta^* = \frac{\theta}{1+\theta}\right),$$

where K is a positive integer. Its probability mass function is given by

$$P(N=n) = \binom{K}{n} (\theta^*)^n (1-\theta^*)^{K-n}, \qquad n = 0,1,\ldots,K. \tag{2.26}$$

The probability generating function of N is

$$A(s) = (1-\theta^*+\theta^* s)^K. \tag{2.27}$$

Consequently, the corresponding long-term survival function is

$$S_p(t) = A(S(t)) = [1-\theta^*+\theta^* S(t)]^K. \tag{2.28}$$

The cure fraction is obtained by taking the limit as $t\to\infty$,

$$p_0 = \lim_{t\to\infty} S_p(t) = (1-\theta^*)^K. \tag{2.29}$$

Thus, increasing K, while keeping θ^* fixed, decreases the cure fraction, since a larger number of potential competing causes increases the probability that at least one cause is present.

Differentiating (2.28) with respect to t, the corresponding density function is

$$f_p(t) = K\theta^* f(t)\,[1-\theta^*+\theta^* S(t)]^{K-1}, \tag{2.30}$$

where $f(t) = -S'(t)$ is the density associated with the proper survival function $S(t)$. Therefore, the corresponding hazard rate function is

$$h_p(t) = \frac{K\theta^* f(t)}{1-\theta^*+\theta^* S(t)}. \tag{2.31}$$

2.4.3 Negative Binomial distribution

A more flexible specification is obtained by assuming that the number of latent competing causes follows a Negative Binomial distribution. Let N follow a Negative Binomial distribution with parameters α and θ, where $\theta > 0$ and $\alpha\theta > -1$. Its probability mass function can be written as

$$P(N=n) = \frac{\Gamma\left(\alpha^{-1}+n\right)}{\Gamma(\alpha^{-1})\,n!}\left(\frac{\alpha\theta}{1+\alpha\theta}\right)^n (1+\alpha\theta)^{-1/\alpha}, \qquad n = 0,1,2,\ldots, \tag{2.32}$$

with the corresponding probability generating function

$$A(s) = [1+\alpha\theta(1-s)]^{-1/\alpha}. \tag{2.33}$$

The long-term survival function is consequently given by

$$S_p(t) = \{1+\alpha\theta\,[1-S(t)]\}^{-1/\alpha}. \tag{2.34}$$

Its cure fraction is

$$p_0 = \lim_{t\to\infty} S_p(t) = (1+\alpha\theta)^{-1/\alpha}. \tag{2.35}$$

Under this parametrization, $E[N] = \theta$ and $\operatorname{Var}(N) = \theta + \alpha\theta$. Therefore, the parameter α controls the dispersion of the number of latent competing causes. For $\alpha > 0$, the distribution is overdispersed relative to a Poisson distribution, whereas $\alpha < 0$ corresponds to underdispersion, subject to the condition $\alpha\theta > -1$.

This parametrization is particularly useful because it provides a continuous connection among the Binomial, Poisson, and Negative Binomial distributions. Specifically, when $\alpha = -1/K$, the Negative Binomial distribution reduces to a Binomial distribution with parameters K and θ/K, provided that $0 \leq \theta/K \leq 1$. As $\alpha \to 0$, the distribution converges to a Poisson distribution with parameter θ. For $\alpha > 0$, it corresponds to the usual overdispersed Negative Binomial formulation.

Differentiating (2.34), the density function associated with the long-term survival distribution is

$$f_p(t) = \theta f(t)\{1+\alpha\theta\,[1-S(t)]\}^{-1/\alpha-1}, \tag{2.36}$$

and its hazard rate function is

$$h_p(t) = \frac{\theta f(t)}{1+\alpha\theta\,[1-S(t)]}. \tag{2.37}$$

The flexibility of this formulation is particularly relevant for long-term survival modeling, since it encompasses several important cure-rate models as special or limiting cases. In particular, the choice $\alpha = -1$ corresponds to the Bernoulli case and consequently recovers the classical mixture cure model, whereas the limit $\alpha \to 0$ yields the Poisson model and the corresponding promotion-time cure model.

2.4.4 Poisson distribution

The Poisson distribution is obtained as the limiting case of the Negative Binomial formulation when $\alpha \to 0$. Let N follow a Poisson distribution with parameter $\theta > 0$, i.e., $N \sim \text{Poisson}(\theta)$. Its probability mass function is

$$P(N = n) = \frac{e^{-\theta}\theta^{n}}{n!}, \qquad n = 0, 1, 2, \ldots, \tag{2.38}$$

and its probability generating function is $A(s) = \exp\{\theta(s-1)\}$. Therefore, the corresponding long-term survival function is

$$S_p(t) = A\left(S(t)\right) = \exp\left\{-\theta[1 - S(t)]\right\} = \exp\left\{-\theta F(t)\right\}. \tag{2.39}$$

The cure fraction is given by $p_0 = \lim_{t\to\infty} S_p(t) = e^{-\theta}$, the corresponding density function is $f_p(t) = \theta f(t) \exp\{-\theta F(t)\}$, and the associated hazard rate function is $h_p(t) = \theta f(t)$. The Poisson specification is particularly important in long-term survival analysis because it leads to the promotion-time cure model (YAKOVLEV; TSODIKOV, 1996).

The Kaplan–Meier estimator (KME) is one of the most widely used nonparametric methods for estimating the survival function in the presence of right-censored observations. In the context of long-term survival analysis, the estimated survival curve can also provide an indication of the presence of a cure fraction when it approaches a positive plateau at large observed times. Although the Kaplan–Meier estimator does not, by itself, identify or consistently estimate the cure fraction without additional assumptions regarding the follow-up period, it provides a useful nonparametric reference for assessing whether a long-term survival pattern is supported by the data. In this work, the Kaplan–Meier estimator is used for this purpose and to assess whether the proposed long-term survival models adequately capture the observed survival pattern.

2.5 Kaplan–Meier estimator

The Kaplan–Meier estimator (KME) is one of the most widely used nonparametric methods for estimating the survival function in the presence of right-censored observations (KAPLAN; MEIER, 1958). Let $\{(t_i, \delta_i)\}_{i=1}^{n}$ denote the observed survival data, where t_i is the observed time and δ_i is the censoring indicator, with $\delta_i = 1$ indicating an observed event and $\delta_i = 0$ indicating right censoring. Let $t_{(1)} < \cdots < t_{(m)}$ denote the distinct observed event times. If d_j individuals experience the event at $t_{(j)}$ and Y_j individuals are at risk immediately before $t_{(j)}$, the KME is given by

$$\widehat{S}_{KM}(t) = \prod_{t_{(j)} \le t} \left(1 - \frac{d_j}{Y_j}\right). \tag{2.40}$$

The KME is a step function that changes only at observed event times, while censored observations contribute to the risk set up to their censoring times. In the context of long-term survival analysis, a survival curve that approaches a positive plateau at large follow-up times may provide empirical evidence of a nonzero long-term survivor fraction. However, such a plateau should be interpreted with caution, as limited follow-up and heavy censoring can also produce a positive tail.

In this work, the KME is used as a nonparametric reference for assessing the long-term survival pattern in the data and for evaluating whether the proposed long-term survival models adequately capture the observed cure fraction.

CHAPTER

3

FOUNDATIONS OF MACHINE LEARNING

3.1 Introduction

3.2 Decision Trees

Decision trees can be used for both regression and classification. They provide the foundation for the tree-based models considered in this work, with particular emphasis on regression trees[1]. The construction of a tree is based on the recursive partitioning of the predictor space, where each partition defines a node and the final resulting partitions are referred to as leaves or terminal nodes. At each internal node, a splitting condition is defined based on one or more predictors. If the condition is satisfied, the observation is directed to one of the resulting child nodes; otherwise, it is directed to the other child node. This process is repeated recursively until the terminal nodes are reached. An example of a regression tree is presented in Figure 3.

For a regression problem, the predictor space is partitioned into J distinct and disjoint regions, denoted by $R_1, R_2, \ldots, R_J$. These regions are typically constructed as rectangular or box-shaped regions through recursive binary splitting, with the objective of minimizing the residual sum of squares. The response variable can then be modeled as a constant c_j within each region R_j. Thus, the resulting regression tree can be represented by

$$f(x) = \sum_{j=1}^{J} c_j I\left(x \in R_j\right). \tag{3.1}$$

The estimator of the constant c_j is obtained by the method of least squares. Thus, for a given region R_j, c_j is chosen to minimize $\sum_{x_i \in R_j} [y_i - f(x_i)]^2$. Since $x_i \in R_j$, the indicator

[1] A regression tree is a particular type of decision tree designed for regression problems.

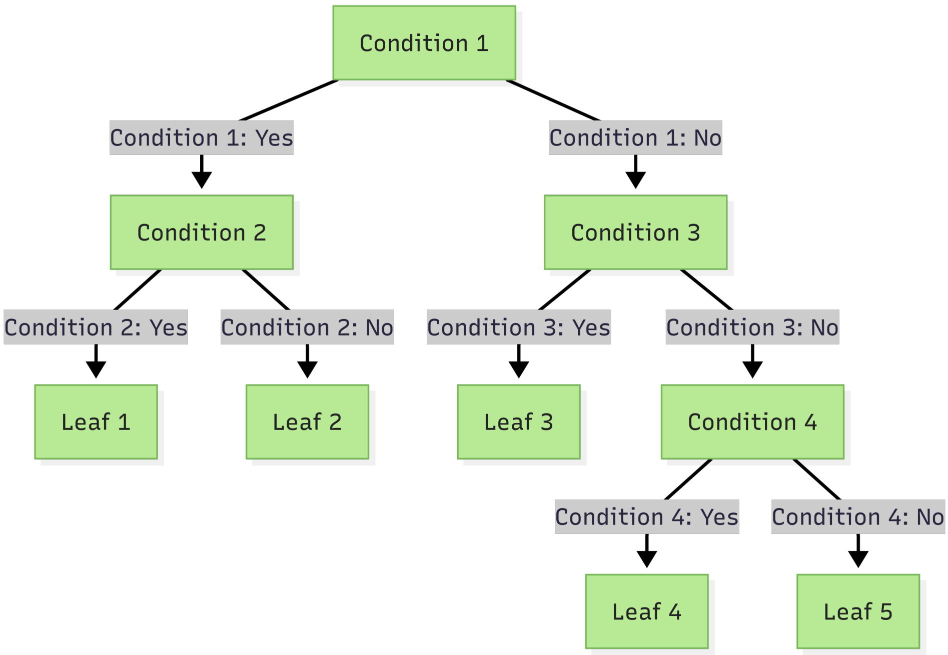

Figure 3 – Example of a regression tree. The tree has five terminal nodes and four internal nodes.

function associated with R_j satisfies

$$I_{R_j}(x_i) = \begin{cases} 1, & \text{if } x_i \in R_j, \\ 0, & \text{if } x_i \notin R_j. \end{cases} \tag{3.2}$$

Because the regions are disjoint, each observation belongs to at most one region. Therefore, for $x_i \in R_j$, the representation in Equation (3.1) reduces to $f(x_i) = c_j$. Consequently, the least-squares problem within region R_j reduces to minimizing

$$\sum_{x_i \in R_j} (y_i - c_j)^2 \tag{3.3}$$

with respect to c_j. Differentiating with respect to c_j gives

$$\frac{\partial}{\partial c_j} \sum_{x_i \in R_j} (y_i - c_j)^2 = -2 \sum_{x_i \in R_j} (y_i - c_j). \tag{3.4}$$

Setting Equation (3.4) equal to zero yields

$$\sum_{x_i \in R_j} (y_i - \hat{c}_j) = 0. \tag{3.5}$$

Expanding the summation and denoting by N_j the number of observations in region R_j, it follows that the estimator of c_j, denoted by $\hat{c}_j$, is simply the sample mean of the observed responses y_i within R_j:

$$\sum_{x_i \in R_j} y_i - \hat{c}_j N_j = 0 \quad \Rightarrow \quad \hat{c}_j = \frac{1}{N_j} \sum_{x_i \in R_j} y_i. \tag{3.6}$$

However, as noted by James *et al.* (2021), considering all possible partitions of the predictor space into J regions is computationally infeasible due to the large number of possible partitions. Therefore, a recursive binary splitting procedure is adopted. The process begins at the root node, which contains all observations, and successively partitions the predictor space into increasingly smaller regions. Each split produces two child nodes, as illustrated in Figure 3.

To perform recursive binary splitting, one first selects a predictor X_j and a cutpoint s such that the resulting partition produces the largest possible reduction in the residual sum of squares. For a given predictor and cutpoint, the predictor space is divided into the two regions

$$R_1(j,s) = \{x : x_j \leq s\} \quad \text{and} \quad R_2(j,s) = \{x : x_j > s\}. \tag{3.7}$$

The predictor j and cutpoint s are then selected by solving

$$\min_{j,s}\left[\min_{c_1}\sum_{x_i \in R_1(j,s)}(y_i - c_1)^2 + \min_{c_2}\sum_{x_i \in R_2(j,s)}(y_i - c_2)^2\right], \tag{3.8}$$

where c_1 and c_2 are the mean responses in the regions $R_1(j,s)$ and $R_2(j,s)$, respectively. Once the optimal split is determined, the observations are partitioned into the two resulting regions, and the same procedure is recursively applied to each subregion.

The size of a regression tree controls the complexity of the resulting model. An excessively large tree may overfit the training data by capturing not only relevant patterns but also random noise. Consequently, although such a model may achieve low prediction error on the training data, its performance on new observations may deteriorate due to poor generalization. Conversely, a tree that is too small may fail to capture important patterns and relationships present in the data. A common strategy for controlling tree complexity is therefore to first grow a large tree T_0, stopping the splitting process only when a specified minimum number of observations is reached in each terminal node. The resulting tree is then pruned according to the cost-complexity criterion, defined below.

Consider a tree T obtained by pruning T_0, such that $T \subseteq T_0$. Let N_j denote the number of observations in region R_j. The residual sum of squares within region R_j can be expressed in terms of the mean squared error

$$Q_j(T) = \frac{1}{N_j}\sum_{x_i \in R_j}\left(y_i - \hat{c}_j\right)^2. \tag{3.9}$$

The cost-complexity criterion is then defined as

$$C_\alpha(T) = \sum_{j=1}^{|T|} N_j Q_j(T) + \alpha|T|, \tag{3.10}$$

where $|T|$ denotes the number of terminal nodes of the tree and $\alpha \geq 0$ is a complexity parameter that controls the trade-off between the goodness of fit and the size of the tree. For each value of

α, the optimal subtree is defined as

$$T_\alpha = \underset{T \subseteq T_0}{\operatorname{argmin}}\, C_\alpha(T). \tag{3.11}$$

Larger values of α impose a greater penalty on tree complexity and therefore favor smaller trees, whereas smaller values of α favor larger trees. When $\alpha = 0$, the complexity penalty vanishes and the resulting tree is T_0.

The sequence of subtrees can be obtained through a weakest-link pruning procedure, in which internal nodes are successively collapsed according to the increase in the residual sum of squares per terminal node removed. This procedure produces a finite sequence of nested subtrees, from which the optimal tree can be selected for a given value of α.

The complexity parameter α can be selected using K-fold cross- validation, commonly with $K = 5$ or $K = 10$. The final regression tree is then given by $T_{\hat{\alpha}}$. The overall procedure for constructing and pruning a regression tree is summarized in Algorithm 1.

Algorithm 1 – Construction and pruning of a regression tree (JAMES *et al.*, 2021).

1. Use recursive binary splitting to grow a large tree T_0 on the training data, stopping only when each terminal node contains fewer than a specified minimum number of observations.
2. Apply the cost-complexity criterion to T_0 to obtain a sequence of optimal subtrees T_α for different values of α.
3. Use K-fold cross-validation to select α. Partition the training observations into K folds. For each $k = 1, \ldots, K$:
    (a) Repeat Steps 1 and 2 using all folds except the k-th fold.
    (b) Evaluate the mean squared prediction error on the held-out k-th fold for each candidate value of α.
    Average the validation errors across the K folds for each value of α, and select $\hat{\alpha}$ that minimizes the average error.
4. Return the subtree $T_{\hat{\alpha}}$ corresponding to the selected value $\hat{\alpha}$.

For classification trees, the main differences from regression trees lie in the criterion used to evaluate node splits and in the way predictions are assigned to terminal nodes. For a classification tree, consider a node j corresponding to a region R_j containing N_j observations. The predicted class is typically the majority class within that node. The proportion of observations belonging to class k is given by

$$\hat{p}_{jk} = \frac{1}{N_j} \sum_{x_i \in R_j} I(y_i = k). \tag{3.12}$$

Thus, each observation in node j is assigned to the class

$$k(j) = \underset{k}{\operatorname{argmax}}\, \hat{p}_{jk}, \tag{3.13}$$

corresponding to the class with the largest estimated proportion in that node.

While the residual sum of squares is used as the impurity measure for regression trees, several measures can be used for classification trees. Common choices include the classification error rate, the Gini index, and cross-entropy. These measures quantify the degree of heterogeneity among the classes within a node and are used to evaluate candidate splits.

3.3 Ensemble Methods

Decision trees offer an interpretable and flexible approach to predictive modeling. However, individual trees can be sensitive to variations in the training data, leading to substantial variability in their predictions. Ensemble learning provides a general framework for addressing this limitation by combining multiple base learners into a single predictive model.

According to Hastie *et al.* (2009), ensemble methods can be viewed as involving two main steps. First, a collection of base learners is constructed from the training data, potentially using different samples, features, or learning procedures. Second, the predictions from these learners are combined to produce an aggregate estimator. The resulting model can achieve improved predictive performance and stability relative to individual base learners. The following sections describe the ensemble methods considered in this work.

3.3.1 Bagging

Bootstrap aggregation, commonly referred to as Bagging, was introduced by Breiman (1996). Its main idea is to construct an aggregate estimator by fitting multiple versions of a base learner to bootstrap samples drawn from the training data. The predictions from these learners are subsequently combined to obtain the final prediction. By averaging multiple fitted models, Bagging can reduce the variance of unstable learning algorithms and consequently improve the stability and predictive performance of the resulting model. In particular, Bagging can be applied to regression trees to mitigate the high variability associated with individual trees.

More formally, let $\mathscr{L}$ denote the training set and let $\{\mathscr{L}^{(i)}\}_{i=1}^{B}$ denote B bootstrap samples drawn independently from $\mathscr{L}$. For each bootstrap sample, a base learner f is fitted, resulting in the collection $f\left(x, \mathscr{L}^{(1)}\right), \ldots, f\left(x, \mathscr{L}^{(B)}\right)$. When the response variable y is continuous, the Bagging estimator is obtained by averaging the predictions across the B fitted models:

$$f_B(x) = \frac{1}{B} \sum_{b=1}^{B} f\left(x, \mathscr{L}^{(b)}\right). \tag{3.14}$$

Here, $f_B(x)$ denotes the aggregated prediction at x.

For classification problems, the predictions are instead combined through majority voting. Let the possible classes be indexed by $j \in \{1, \ldots, J\}$, and define

$$N_j(x) = \sum_{b=1}^{B} I\left\{f\left(x, \mathscr{L}^{(b)}\right) = j\right\}, \tag{3.15}$$

where $N_j(x)$ denotes the number of bootstrap models that assign observation x to class j. The Bagging classification rule is then given by

$$f_B(x) = \underset{j \in \{1,\ldots,J\}}{\operatorname{argmax}} \; N_j(x). \tag{3.16}$$

Thus, the predicted class is the one receiving the largest number of votes among the B fitted base learners.

3.4 Gradient Boost

3.5 Kernel Methods

Kernel-based methods are highly flexible techniques that can be used to extend models or algorithms to nonlinear decision regions (MOHRI; ROSTAMIZADEH; TALWALKAR, 2018). These methods are based on kernel functions, which, under symmetry and positive semidefiniteness conditions, can be interpreted as inner products in high-dimensional feature spaces. In particular, positive semidefinite kernels are associated with Reproducing Kernel Hilbert Spaces (RKHS). Therefore, this section aims to introduce the fundamental concepts of kernel-based methods, including the definition of kernel functions, positive semidefinite kernels, the representer theorem, and Reproducing Kernel Hilbert Spaces (RKHS).

3.6 Kernels

By definition, a function $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ is called a kernel on $\mathscr{X}$. A central idea is to construct a kernel K such that, for any two points $x, x' \in \mathscr{X}$, $K(x,x')$ corresponds to the inner product between representations of these points in a Hilbert space $\mathscr{H}$, that is,

$$\forall x, x' \in \mathscr{X}, \quad K(x,x') = \langle \phi(x), \phi(x') \rangle_{\mathscr{H}}, \tag{3.17}$$

for some mapping $\phi : \mathscr{X} \to \mathscr{H}$. The space $\mathscr{H}$ is referred to as the feature space.

From this perspective, the kernel can be interpreted as a measure of similarity between elements of $\mathscr{X}$, since the inner product provides a notion of similarity between their representations in the feature space.

A fundamental advantage of this approach is computational. In many cases, it is significantly more efficient to compute $K(x,x')$ directly than to explicitly compute $\phi(x)$ and $\phi(x')$ and then evaluate their inner product. This idea is commonly referred to as the *kernel trick*.

3.6.1 *Positive Semidefinite Kernels*

For a function K to be used as a positive semidefinite kernel, it must satisfy the appropriate symmetry and positive semidefiniteness conditions. These properties allow the kernel to induce a consistent inner-product structure and, consequently, a geometry in a suitable Hilbert space.

Formally, a kernel $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ is said to be positive semidefinite if, for any finite set $\{x_1, \ldots, x_n\} \subset \mathscr{X}$, the Gram matrix

$$\mathbf{K} = [K(x_i, x_j)]_{i,j=1}^{n} \in \mathbb{R}^{n \times n} \tag{3.18}$$

is positive semidefinite. That is, for any vector $\mathbf{c} = (c_1, \ldots, c_n)^\top \in \mathbb{R}^n$,

$$\mathbf{c}^\top \mathbf{K} \mathbf{c} = \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j K(x_i, x_j) \geq 0. \tag{3.19}$$

Equivalently, all eigenvalues of $\mathbf{K}$ are nonnegative. This property is essential because it guarantees that the kernel admits a consistent inner-product representation in a Hilbert space.

3.6.2 Hilbert Spaces

The notion of a Hilbert space provides the mathematical structure required for the theory of kernels.

By definition, a Hilbert space $\mathscr{H}$ is a vector space equipped with an inner product $\langle \cdot, \cdot \rangle_{\mathscr{H}}$ that is complete with respect to the norm induced by this inner product. The norm on $\mathscr{H}$ is defined by

$$\forall h \in \mathscr{H}, \quad \|h\|_{\mathscr{H}} = \sqrt{\langle h, h \rangle_{\mathscr{H}}}. \tag{3.20}$$

Completeness implies that every Cauchy sequence in $\mathscr{H}$ converges to an element that also belongs to $\mathscr{H}$. This property is important in approximation and optimization problems, as it ensures that limits of convergent sequences remain within the space.

In the context of kernel methods, the Hilbert space provides the space in which data can be represented through a feature mapping ϕ.

3.6.3 Reproducing Kernel Hilbert Spaces

The relationship between positive semidefinite kernels and Hilbert spaces can be formalized through the theory of Reproducing Kernel Hilbert Spaces (RKHS).

Let $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ be a symmetric positive semidefinite kernel. Then, there exists a Hilbert space $\mathscr{H}$ of functions on $\mathscr{X}$ such that, for every $x, x' \in \mathscr{X}$,

$$K(x, x') = \langle K(x, \cdot), K(x', \cdot) \rangle_{\mathscr{H}}. \tag{3.21}$$

Moreover, $\mathscr{H}$ satisfies the so-called reproducing property:

$$\forall h \in \mathscr{H}, \forall x \in \mathscr{X}, \quad h(x) = \langle h, K(x, \cdot) \rangle_{\mathscr{H}}. \tag{3.22}$$

The reproducing property implies that the evaluation functional $h \mapsto h(x)$ is continuous for every $x \in \mathscr{X}$ and can be represented by the function $K(x,\cdot)$ through the Riesz representation theorem.

A Hilbert space satisfying these properties is called a Reproducing Kernel Hilbert Space (RKHS) associated with the kernel K. RKHSs play a central role in statistical learning methods, particularly in regularization, nonparametric estimation, and supervised learning.

3.6.4 Representer Theorem

One of the most important results for the formulation of the GG-KM model is the Representer Theorem. Its main importance lies in the fact that, although the optimization is performed over a potentially infinite-dimensional space, an optimal solution can be expressed as a finite linear combination of kernels evaluated at the observations in the sample.

Formally, let $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ be a positive semidefinite kernel and let $\mathscr{H}$ be its associated RKHS. For any nondecreasing function $G : \mathbb{R} \to \mathbb{R}$ and any loss function $L : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$, consider the optimization problem

$$\min_{h \in \mathscr{H}} \left\{ G\left(\|h\|_{\mathscr{H}}\right) + L\left(h(x_1), \ldots, h(x_n)\right) \right\}. \tag{3.23}$$

Under the appropriate conditions for the existence of a minimizer, this problem admits a solution of the form

$$h^*(\cdot) = \sum_{i=1}^{n} \alpha_i K(x_i, \cdot). \tag{3.24}$$

If G is strictly increasing, then every minimizer admits this representation.

This result is particularly relevant because it reduces a functional optimization problem in an infinite-dimensional space to a finite-dimensional optimization problem involving only the coefficients $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)^\top$. In practice, this makes kernel-based methods computationally feasible and shows that the necessary information for estimation can be obtained from the kernel matrix, without explicitly computing the feature mapping $\phi(\cdot)$.

CHAPTER

4

PROPOSED METHODS

4.1 Introduction

4.2 GG-KM

The GG-KM (an acronym for Generalized Gamma Distribution and Kernel Machines) is a novel semiparametric approach to long-term survival modeling. The main objective is to address the limitations of linear predictors by employing kernel functions to model $\log(\theta(\mathbf{x}))$ within a Reproducing Kernel Hilbert Space (RKHS), where $\theta(\mathbf{x})$ is a parameter governing the distribution of the latent number of competing causes in the promotion cure model. This formulation allows for a flexible, potentially nonlinear relationship between the covariates and the latent competing-risk structure.

To provide greater flexibility in modeling the survival distribution of susceptible individuals, the Generalized Gamma distribution, introduced by Stacy (1962) is adopted for the latent event times. This distribution includes several commonly used parametric survival distributions as special cases and, consequently, allows a broad range of hazard shapes to be represented. Its probability density and cumulative density functions are defined as:

$$f(t;a,d,p) = \frac{pt^{d-1}\exp\left[-(t/a)^p\right]}{a^d\Gamma(d/p)}, \qquad F(t;a,d,p) = \frac{\gamma(d/p,(t/a)^p)}{\Gamma(d/p)} \tag{3}$$

where $a > 0$ is the scale parameter, and d, $p > 0$ are the shape parameters. The Generalized Gamma distribution is a powerful generalization. For example, if $d = p$, it reduces to the Weibull distribution. Also, if $p = 1$ or $d/p \to \infty$ it reduces to the Standard Gamma and converges to the Log-Normal distribution, respectively.

The model is estimated through an Expectation–Maximization (EM) (MCLACHLAN; KRISHNAN, 2008) algorithm and is formulated for different probability distributions of the latent number of competing causes N, namely the Bernoulli, Binomial, Poisson, and Negative

Binomial distributions. This formulation provides a unified framework for investigating different assumptions regarding the latent competing-risk structure while preserving the kernel-based semiparametric formulation.

4.2.1 Nonparametric extension via kernel methods

Although the classical promotion-time cure model allows covariates to be incorporated through parametric functions of $\theta(\mathbf{x})$, such an approach may be restrictive when the relationship between the covariates and the latent competing-risk structure is complex or nonlinear.

In this context, the proposition is to model the parameter governing the distribution of the latent competing causes, $\theta(\mathbf{x})$, through a nonparametric function

$$g(\mathbf{x}) = \log(\theta(\mathbf{x})), \tag{4.1}$$

where $g : \mathscr{X} \to \mathbb{R}$. The logarithmic transformation ensures that $\theta(\mathbf{x}) > 0$ for every $\mathbf{x}$. The estimation of g is formulated through the minimization of a penalized negative log-likelihood:

$$\min_{g \in \mathscr{H}_K} \left\{ -\sum_{i=1}^{n} l_i(g) + \frac{\lambda}{2} \|g\|^2_{\mathscr{H}_K} \right\}, \tag{4.2}$$

where $l_i(g)$ denotes the log-likelihood contribution of the ith individual, $\mathscr{H}_K$ is a Reproducing Kernel Hilbert Space (RKHS) associated with a positive-definite kernel $K(\cdot,\cdot)$, and $\lambda \geq 0$ is a regularization parameter controlling the complexity of the estimated function.

By the Representer Theorem, the solution to the optimization problem in Equation (4.2) admits a finite-dimensional representation of the form

$$g^*(\mathbf{x}) = \sum_{i=1}^{n} \alpha_i K(\mathbf{x}, \mathbf{x}_i), \tag{4.3}$$

where $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ are coefficients to be estimated. Consequently, the parameter governing the distribution of the latent competing causes for an individual with covariate vector $\mathbf{x}_i$ is given by

$$\theta(\mathbf{x}_i) = \exp\left\{ \sum_{j=1}^{n} \alpha_j K(\mathbf{x}_i, \mathbf{x}_j) \right\}. \tag{4.4}$$

4.2.2 Estimation Procedures

The estimation of the proposed models is performed using the Expectation–Maximization (EM) algorithm. Since the number of latent competing causes N_i is unobserved, direct maximization of the observed-data likelihood may be analytically and computationally challenging. The EM algorithm addresses this difficulty by treating N_i, $i = 1, \dots, n$, as latent variables and iteratively maximizing the expected complete-data log-likelihood.

Table 2 – Complete-data log-likelihoods and conditional expectations of the latent number of competing causes for the distributions considered for N_i.

Distribution of N_i	Complete-data log-likelihood ℓ_c	$\widehat{N_i} = E(N_i \mid t_i, \delta_i)$
Bernoulli(θ_i)	$\sum_{i=1}^{n} \Big[n_i \log \theta_i + (1-n_i)\log(1-\theta_i) + \delta_i \log n_i + \delta_i \log f_i + (n_i - \delta_i)\log S_i \Big]$	$\delta_i + (1-\delta_i)\dfrac{\theta_i S_i}{1-\theta_i+\theta_i S_i}$
Binomial(K, θ_i^*)	$\sum_{i=1}^{n} \Big[\log \binom{K}{n_i} + n_i \log \theta_i^* + (K-n_i)\log(1-\theta_i^*) + \delta_i \log n_i + \delta_i \log f_i + (n_i-\delta_i)\log S_i \Big]$	$\delta_i + (K-\delta)\dfrac{\theta_i^* S_i}{1-\theta_i^*+\theta_i^* S_i}$
Negative Binomial(α, θ_i)	$\sum_{i=1}^{n} \Big[\log\Gamma(\alpha^{-1}+n_i) - \log\Gamma(\alpha^{-1}) - \log(n_i!) + n_i \log\left(\dfrac{\alpha\theta_i}{1+\alpha\theta_i}\right) - \dfrac{1}{\alpha}\log(1+\alpha\theta_i) + \delta_i \log n_i + \delta_i \log f_i + (n_i-\delta_i)\log S_i \Big]$	$\delta_i + \dfrac{(1+\alpha\delta_i)\,\theta_i S_i}{1+\alpha\theta_i F_i}$
Poisson(θ_i)	$\sum_{i=1}^{n} \Big[n_i \log \theta_i - \log(n_i!) - \theta_i + \delta_i \log n_i + \delta_i \log f_i + (n_i-\delta_i)\log S_i \Big]$	$\delta_i + \theta_i S_i$

Notes: For compactness, $S_i = S(t_i)$ and $f_i = f(t_i)$. For the The conditional expectations are evaluated at the observed pair (t_i, δ_i) and correspond to the E-step of the EM algorithm.

Let $\mathscr{D}_{\text{obs}} = \{(t_i, \delta_i, \mathbf{x}_i)\}_{i=1}^{n}$ denote the observed data and let $\mathscr{D}_{\text{comp}} = \{(t_i, \delta_i, n_i, \mathbf{x}_i)\}_{i=1}^{n}$ denote the corresponding complete data. Let $\boldsymbol{\eta}$ denote the vector of model parameters. The observed-data likelihood is obtained by integrating out the latent variables:

$$L_{\text{obs}}\left(\boldsymbol{\eta}; \mathscr{D}_{\text{obs}}\right) = \sum_{\boldsymbol{n}} L_c\left(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}\right), \tag{4.5}$$

where the summation is taken over all possible realizations of $\boldsymbol{n} = (n_1, \ldots, n_n)$ according to the support of the corresponding distribution of N_i.

The complete-data log-likelihood can be shown to be Table 2 and their proof are described in Appendix A.1. These results are used as teh basis for the EM procedure. Given a current parameter estimate $\boldsymbol{\eta}^{(r)}$, the E-step consists of computing the conditional expectation

$$Q\left(\boldsymbol{\eta} \mid \boldsymbol{\eta}^{(r)}\right) = \mathbb{E}_{\boldsymbol{\eta}^{(r)}}\left[\ell_c\left(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}\right) \mid \mathscr{D}_{\text{obs}}\right]. \tag{4.6}$$

In particular, the expectation is taken with respect to the conditional distribution of the latent number of competing causes,

$$P_{\boldsymbol{\eta}^{(r)}}\left(N_i = n_i \mid t_i, \delta_i, \mathbf{x}_i\right).$$

Consequently, the quantities involving N_i in the complete-data log-likelihood are replaced by their corresponding conditional expectations under the current parameter estimates.

In the M-step, the expected complete-data log-likelihood is maximized with respect to the model parameters $\boldsymbol{\eta}^{(r+1)} = \arg\max_{\boldsymbol{\eta}} Q\left(\boldsymbol{\eta} \mid \boldsymbol{\eta}^{(r)}\right)$.

Because the proposed GG-KM model estimates the nonparametric function $g(\mathbf{x})$ in an RKHS, regularization is incorporated into the M-step. Thus, the corresponding Penalized EM algorithm maximizes $Q_\lambda\left(\boldsymbol{\eta} \mid \boldsymbol{\eta}^{(r)}\right) = Q\left(\boldsymbol{\eta} \mid \boldsymbol{\eta}^{(r)}\right) - \frac{\lambda}{2}\|f\|^2_{\mathscr{H}_K}$. The resulting update is therefore $\boldsymbol{\eta}^{(r+1)} = \arg\max_{\boldsymbol{\eta}} Q_\lambda\left(\boldsymbol{\eta} \mid \boldsymbol{\eta}^{(r)}\right)$.

For the Bernoulli, Binomial, and Poisson specifications, the resulting optimization problems can be handled directly within this penalized EM framework. However, the Negative Binomial formulation introduces the additional dispersion parameter α. In this case, the expected complete-data log-likelihood contains terms involving the Gamma function and α, which lead to a considerably more complicated optimization problem. In particular, the expectation required for the update of α does not generally admit a convenient closed-form expression and may be computationally expensive to evaluate repeatedly during the M-step.

To address this difficulty, the ECME (Expectation/Conditional Maximization Either) algorithm is considered for the Negative Binomial model (MCLACHLAN; KRISHNAN, 2008). ECME is an extension of the EM framework in which some parameter blocks are updated by maximizing the expected complete-data log-likelihood, as in the conventional EM algorithm, while other parameter blocks may instead be updated by directly maximizing the observed-data log-likelihood.

Thus, for the Negative Binomial model, the ECME procedure is used specifically for the estimation of the dispersion parameter α. Rather than requiring the maximization of a complicated expected complete-data contribution with respect to α, the observed-data log-likelihood can be used for this parameter block. The remaining parameters are updated through the usual penalized conditional maximization steps. This strategy preserves the EM framework while avoiding an unnecessarily complicated computation in the estimation of α.

Algorithm 2 – Algoritmo EM penalizado para estimar o modelo GG-KM - $N_i \sim \text{Poisson}\big(\theta(\mathbf{x}_i)\big)$.

Inicialize $\alpha^{(0)}$, $a^{(0)}$, $d^{(0)}$, $p^{(0)}$ e escolha uma tolerância $\varepsilon > 0$.
Defina $k \leftarrow 0$.
repeat
    E-step: para $i = 1, \ldots, n$, calcule

$$\widehat{N}_i^{(k+1)} = \delta_i + \exp\left(\sum_{j=1}^{n} \alpha_j^{(k)} K(x_i, x_j)\right) S_{GG}\Big(t_i; a^{(k)}, d^{(k)}, p^{(k)}\Big).$$

    M-step:

$$\alpha^{(k+1)} = \arg\max_{\alpha} \left\{ \sum_{i=1}^{n} \left[-\exp\left(\sum_{j=1}^{n} \alpha_j K(x_i, x_j)\right) + \widehat{N}_i^{(k+1)} \sum_{j=1}^{n} \alpha_j K(x_i, x_j) \right] - \frac{\lambda}{2} \alpha^\top K \alpha \right\}.$$

$$(a^{(k+1)}, d^{(k+1)}, p^{(k+1)}) = \arg\max_{a,d,p} \sum_{i=1}^{n} \Big[\delta_i \log f_{GG}(t_i; a, d, p) + (\widehat{N}_i^{(k+1)} - \delta_i) \log S_{GG}(t_i; a, d, p) \Big].$$

    Calcule a log-verossimilhança penalizada observada

$$\ell_P^{(k+1)} = \ell_P\Big(\alpha^{(k+1)}, a^{(k+1)}, d^{(k+1)}, p^{(k+1)}\Big).$$

    Atualize

$$\Delta^{(k+1)} = \left| \ell_P^{(k+1)} - \ell_P^{(k)} \right|.$$

    $k \leftarrow k + 1$.
until $\Delta^{(k+1)} < \varepsilon$

Algorithm 3 – Algoritmo ECME para estimar o modelo GG-KM com riscos latentes Binomial Negativa

1: Inicialize $\alpha^{(0)}$, $a^{(0)}$, $d^{(0)}$, $p^{(0)}$, $\alpha^{(0)}$ e escolha uma tolerância $\varepsilon > 0$.

2: Defina $k \leftarrow 0$.

3: **repeat**

4: Para $i = 1, \ldots, n$, calcule

$$\theta_i^{(k)} = \exp\left(\sum_{j=1}^{n} \alpha_j^{(k)} K(x_i, x_j)\right),$$

$$F_{GG}^{(k)}(t_i) = F_{GG}\left(t_i; a^{(k)}, d^{(k)}, p^{(k)}\right), \qquad S_{GG}^{(k)}(t_i) = 1 - F_{GG}^{(k)}(t_i).$$

5: **E-step:**

$$\widehat{N}_i^{(k+1)} = \delta_i + \frac{\left(1 + \alpha^{(k)}\delta_i\right)\theta_i^{(k)} S_{GG}^{(k)}(t_i)}{1 + \alpha^{(k)}\theta_i^{(k)} F_{GG}^{(k)}(t_i)}, \qquad i = 1, \ldots, n.$$

6: **M-step:**

$$\alpha^{(k+1)} = \arg\max_{\alpha} \left\{ \sum_{i=1}^{n} \left[\widehat{N}_i^{(k+1)} \log(\theta_i) - \left(\widehat{N}_i^{(k+1)} + \frac{1}{\alpha^{(k)}} \right) \log\left(1 + \alpha^{(k)}\theta_i\right) \right] - \frac{\lambda}{2} \alpha^{\top} K \alpha \right\},$$

$$(a^{(k+1)}, d^{(k+1)}, p^{(k+1)}) = \arg\max_{a,d,p} \sum_{i=1}^{n} \left[\delta_i \log f_{GG}(t_i; a, d, p) + (\widehat{N}_i^{(k+1)} - \delta_i) \log S_{GG}(t_i; a, d, p) \right].$$

$$\alpha^{(k+1)} = \arg\max_{\alpha > 0} \ell_O\left(\alpha^{(k+1)}, a^{(k+1)}, d^{(k+1)}, p^{(k+1)}, \alpha\right).$$

7: Calcule a log-verossimilhança penalizada observada

$$\ell_P^{(k+1)} = \ell_P\left(\alpha^{(k+1)}, a^{(k+1)}, d^{(k+1)}, p^{(k+1)}, \alpha^{(k+1)}\right).$$

8: Atualize

$$\Delta^{(k+1)} = \left| \ell_P^{(k+1)} - \ell_P^{(k)} \right|.$$

9: $k \leftarrow k + 1$.

10: **until** $\Delta^{(k+1)} < \varepsilon$

Algorithm 4 – Algoritmo EM penalizado para estimar o modelo GG-KM - $N_i \sim \text{Binomial}(K, \theta_i^*)$.

Inicialize $\alpha^{(0)}, a^{(0)}, d^{(0)}, p^{(0)}$ e escolha uma tolerância $\varepsilon > 0$.

Defina $k \leftarrow 0$.

repeat

E-step: para $i = 1, \ldots, n$, calcule

$$f_i^{(k)} = \sum_{j=1}^{n} \alpha_j^{(k)} K(x_i, x_j),$$

$$\theta_i^{*(k)} = \frac{\exp(f_i^{(k)})}{1 + \exp(f_i^{(k)})},$$

e

$$\widehat{N}_i^{(k+1)} = \delta_i + (K - \delta_i) \frac{\theta_i^{*(k)} S_{GG}\left(t_i; a^{(k)}, d^{(k)}, p^{(k)}\right)}{1 - \theta_i^{*(k)} + \theta_i^{*(k)} S_{GG}\left(t_i; a^{(k)}, d^{(k)}, p^{(k)}\right)}.$$

M-step:

Atualize α resolvendo

$$\alpha^{(k+1)} = \arg\max_{\alpha} \left\{ \sum_{i=1}^{n} \left[\widehat{N}_i^{(k+1)} f_i - K \log\left(1 + \exp(f_i)\right) \right] - \frac{\lambda}{2} \alpha^\top K \alpha \right\},$$

onde

$$f_i = \sum_{j=1}^{n} \alpha_j K(x_i, x_j).$$

Atualize os parâmetros da distribuição Generalizada Gama:

$$(a^{(k+1)}, d^{(k+1)}, p^{(k+1)}) = \arg\max_{a,d,p} \sum_{i=1}^{n} \left[\delta_i \log f_{GG}(t_i; a, d, p) + (\widehat{N}_i^{(k+1)} - \delta_i) \log S_{GG}(t_i; a, d, p) \right].$$

Calcule a log-verossimilhança penalizada observada

$$\ell_P^{(k+1)} = \ell_P\left(\alpha^{(k+1)}, a^{(k+1)}, d^{(k+1)}, p^{(k+1)}\right).$$

Atualize

$$\Delta^{(k+1)} = \left| \ell_P^{(k+1)} - \ell_P^{(k)} \right|.$$

$k \leftarrow k + 1$.

until $\Delta^{(k+1)} < \varepsilon$

Algorithm 5 – EM with functional gradient boosting for the GG-KM model - $N_i \sim \text{Binomial}(K, \theta_i^*)$.

1: Initialize $f^{(0)}(\cdot), a^{(0)}, d^{(0)}, p^{(0)}$ and choose $\varepsilon > 0$, $\varepsilon_{\text{tree}} > 0$, and the maximum number of trees M.

2: Set $k \leftarrow 0$.

3: **repeat**

4: **E-step:** compute, for $i = 1, \ldots, n$,

$$\theta_i^{*(k)} = \frac{\exp\{f^{(k)}(x_i)\}}{1+\exp\{f^{(k)}(x_i)\}},$$

$$\widehat{N}_i^{(k+1)} = \delta_i + (K - \delta_i)\frac{\theta_i^{*(k)} S_{GG}(t_i; a^{(k)}, d^{(k)}, p^{(k)})}{1 - \theta_i^{*(k)} + \theta_i^{*(k)} S_{GG}(t_i; a^{(k)}, d^{(k)}, p^{(k)})}.$$

5: **M-step for $f(x)$:** set $f_0(x) \leftarrow f^{(k)}(x)$ and, for $m = 1, \ldots, M$, compute

$$g_{i,m} = \widehat{N}_i^{(k+1)} - K\frac{\exp\{f_{m-1}(x_i)\}}{1+\exp\{f_{m-1}(x_i)\}},$$

fit a regression tree $h_m(x)$ to $\{(x_i, g_{i,m})\}_{i=1}^n$, and obtain

$$\rho_m = \arg\max_{\rho \geq 0} \sum_{i=1}^n \left[\widehat{N}_i^{(k+1)}\{f_{m-1}(x_i) + \rho h_m(x_i)\} - K\log\left(1 + \exp\{f_{m-1}(x_i) + \rho h_m(x_i)\}\right)\right].$$

6: Update

$$f_m(x) = f_{m-1}(x) + \rho_m h_m(x),$$

and stop the boosting iterations if

$$\frac{1}{n}\sum_{i=1}^n |f_m(x_i) - f_{m-1}(x_i)| < \varepsilon_{\text{tree}}.$$

7: Set $f^{(k+1)}(x) \leftarrow f_m(x)$.

8: **M-step for the Generalized Gamma parameters:**

$$(a^{(k+1)}, d^{(k+1)}, p^{(k+1)}) = \arg\max_{a,d,p} \sum_{i=1}^n \left[\delta_i \log f_{GG}(t_i; a, d, p) + (\widehat{N}_i^{(k+1)} - \delta_i)\log S_{GG}(t_i; a, d, p)\right].$$

9: Compute

$$\Delta^{(k+1)} = \left|\ell_P^{(k+1)} - \ell_P^{(k)}\right|,$$

where

$$\ell_P^{(k+1)} = \ell_P\left(f^{(k+1)}, a^{(k+1)}, d^{(k+1)}, p^{(k+1)}\right).$$

10: $k \leftarrow k+1$.

11: **until** $\Delta^{(k+1)} < \varepsilon$

CHAPTER 5

EXPERIMENTAL EVALUATION

5.1 Particionamento dos dados

A avaliação do desempenho preditivo dos modelos foi conduzida por meio de um esquema de validação baseado na divisão dos dados em conjuntos de treinamento e teste. Especificamente, adotou-se uma estratégia de holdout, na qual 70% das observações foram utilizadas para treinamento e os 30% restantes reservados para teste.

A separação foi realizada de forma aleatória, preservando, sempre que aplicável, a estrutura dos dados censurados, de modo a garantir que ambos os subconjuntos apresentassem proporções semelhantes de observações censuradas e não censuradas. Essa precaução é particularmente importante em problemas de análise de sobrevivência, uma vez que a distribuição dos tipos de censura pode influenciar significativamente o desempenho dos modelos.

Adicionalmente, com o objetivo de reduzir a variabilidade associada a uma única partição, o procedimento de holdout foi repetido múltiplas vezes. As métricas de desempenho foram então agregadas ao longo dessas repetições.

5.2 Estudo de simulação e Métricas de Erro

Com o objetivo de avaliar o desempenho do modelo GG-KM em diferentes cenários de complexidade da fração de cura, foi conduzido um estudo de simulação, seguindo a mesma estrutura de geração de dados proposta por Pal and Aselisewine (2023), adaptada ao contexto do modelo de tempo de promoção.

Foram considerados dois tamanhos de amostra, $n = 300$ e $n = 600$. Em cada replicação, os dados foram particionados em conjunto de treinamento e teste, sendo dois terços das observações destinados ao treinamento e o terço restante utilizado para avaliação preditiva.

Cada conjunto de dados foi gerado a partir de duas covariáveis independentes,

$$x_1, x_2 \sim N(0,1).$$

A estrutura de cura foi construída a partir de três métodos distintos para a geração da probabilidade de não cura, permitindo avaliar o comportamento do modelo sob diferentes graus de linearidade e não linearidade. Para o primeiro caso, considera-se uma fronteira linear, dada por

$$\pi(\mathbf{x}) = \frac{\exp(0.3 - 5x_1 - 3x_2)}{1 + \exp(0.3 - 5x_1 - 3x_2)}.$$

Esse cenário representa a estrutura clássica de regressão logística, em que indivíduos curados e não curados são linearmente separáveis.

No segundo método, é introduzido uma relação quadrática entre as covariáveis, com o objetivo de avaliar a capacidade do modelo em capturar superfícies de decisão não lineares. Portanto, tem-se:

$$\pi(\mathbf{x}) = \frac{\exp(0.3 + 5x_1^2 - 3x_2^2)}{1 + \exp(0.3 + 5x_1^2 - 3x_2^2)}.$$

Por fim, o para o terceiro método, é considerado uma estrutura não linear e periódica:

$$\pi(\mathbf{x}) = \exp\{-\exp(0.3 - 5\cos(x_1) - 3\sin(x_2))\}.$$

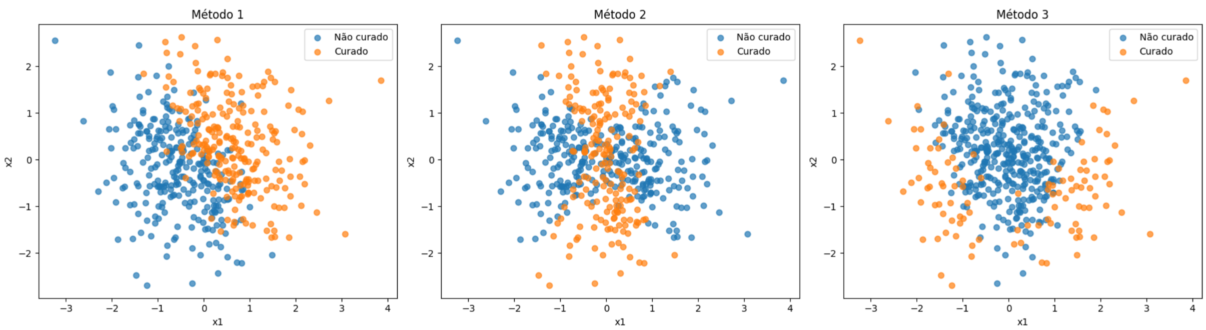

Figure 4 – Simulação de curados e não curados sobre os três métodos descritos. Respectivamente, pode-se observar o método 1, 2 e 3.

Esse cenário impõe fronteiras de classificação complexas, sendo particularmente útil para avaliar a flexibilidade da abordagem baseada em kernels. Os dados simulados, considerando cada um dos três casos, pode ser visualizado na Figura 4.

Para a componente de latência, os tempos associados aos riscos latentes foram gerados a partir de uma distribuição Weibull, cuja função de risco é dada por

$$h(t|\mathbf{x}) = h_0(t)\exp(\beta_1 x_1 + \beta_2 x_2),$$

onde a função de risco basal foi definida como

$$h_0(t) = \alpha t^{\alpha-1}, \quad \alpha > 0.$$

Essa especificação implica que os tempos latentes seguem uma distribuição Weibull com parâmetro de forma α e parâmetro de escala $\lambda(\mathbf{x}) = \{\exp(\beta_1 x_1 + \beta_2 x_2)\}^{-1/\alpha}$.

Os valores utilizados para os parâmetros foram $(\alpha, \beta_1, \beta_2) = (2, 1, 0.5)$ Os tempos de censura foram gerados independentemente a partir de uma distribuição exponencial com taxa igual a 0.2, isto é $C_i \sim \text{Exp}(0.2)$. A partir desta configuração, a proporção de censura obtida para o método 1, 2 e 3 é de aproximadamente 0,60, 0,50 e 0,40, respectivamente. Para a proporção de cura foi obtido aproximadamente 0,50, 0,60 e 0,20.

A geração dos dados simulados foi realizada com base na estrutura do modelo de fração de cura em tempo de promoção. Nesse contexto, cada indivíduo pode pertencer a um de dois grupos: curados ou suscetíveis ao evento de interesse. A probabilidade de um indivíduo ser suscetível é dada por $\pi(\mathbf{x}_i)$, enquanto a probabilidade de ser curado é $1 - \pi(\mathbf{x}_i)$.

O processo de simulação ocorre em duas etapas. Inicialmente, para cada indivíduo, gera-se uma variável uniforme U_i e um tempo de censura C_i. O valor de U_i é utilizado para determinar se o indivíduo pertence ao grupo curado ou suscetível. Caso $U_i \leq 1 - \pi(\mathbf{x}_i)$, o indivíduo é classificado como curado e seu tempo observado corresponde apenas ao tempo de censura.

Por outro lado, se $U_i > 1 - \pi(\mathbf{x}_i)$, o indivíduo é considerado suscetível. Nesse caso, gera-se um tempo latente de falha Y_i utilizando o método da transformada inversa aplicado à função de sobrevivência populacional. Para isso, sorteia-se uma nova variável uniforme no intervalo $(1 - \pi(\mathbf{x}_i), 1)$ e resolve-se a equação associada à função de sobrevivência, obtendo assim o tempo de falha latente.

Por fim, o tempo observado é definido como o mínimo entre o tempo latente de falha e o tempo de censura. O indicador de falha é então determinado de acordo com qual desses tempos ocorreu primeiro. Esse procedimento garante que os dados simulados preservem simultaneamente a estrutura de fração de cura e a presença de censura à direita. O algoritmo pode ser visualizado no Algoritmo 6

Algorithm 6 – Geração de dados simulados com fração de cura

for $i = 1, \dots, n$ **do**
  Gerar $U_i \sim \text{Uniform}(0,1)$
  Gerar $C_i \sim \text{Exp}(0.2)$
  if $U_i \leq 1 - \pi(\mathbf{x}_i)$ **then**
    Definir $t_i \leftarrow C_i$
    Definir $\delta_i \leftarrow 0$
  else
    Gerar $U_i^* \sim \text{Uniform}(1 - \pi(\mathbf{x}_i), 1)$
    Resolver

$$(1 - \pi(\mathbf{x}_i))^{F(Y_i|\mathbf{x}_i)} = U_i^*$$

    Obter

$$Y_i = F^{-1}\left(\frac{\log(U_i^*)}{\log(1 - \pi(\mathbf{x}_i))}\right)$$

    Definir $t_i \leftarrow \min(Y_i, C_i)$
    if $t_i = Y_i$ **then**
      Definir $\delta_i \leftarrow 1$
    else
      Definir $\delta_i \leftarrow 0$
    end if
  end if
end for

O desempenho do modelo GG-KM foi avaliado com base no viés, erro quadrático médio (EQM) e no *Integrated Brier Score* (IBS) com ajuste IPCW (*Inverse Probability of Censoring Weighting*) (DONG *et al.*, 2020), permitindo mensurar tanto a qualidade da estimação dos parâmetros quanto a capacidade preditiva do modelo.

Seja $\hat{S}_i(t)$ a probabilidade de sobrevivência estimada para o indivíduo i no tempo t, e T_i^* o tempo observado. Os pesos IPCW são definidos por

$$\omega_i(t) = \frac{\mathbb{I}(T_i^* \leq t, \delta_i = 1)}{\hat{G}(T_i^*)} + \frac{\mathbb{I}(T_i^* > t)}{\hat{G}(t)},$$

em que $\hat{G}(t)$ representa a função de sobrevivência do mecanismo de censura, geralmente estimada pelo método de Kaplan-Meier.

Com isso, o Brier Score ajustado por IPCW no tempo t é dado por

$$BS_{IPCW}(t) = \frac{1}{\tilde{n}(t)} \sum_{i=1}^{n} \left[\hat{S}_i(t)^2 \frac{\mathbb{I}(T_i^* \leq t, \delta_i = 1)}{\hat{G}(T_i^*)} + (1 - \hat{S}_i(t))^2 \frac{\mathbb{I}(T_i^* > t)}{\hat{G}(t)} \right],$$

em que

$$\tilde{n}(t) = \sum_{i=1}^{n} \omega_i(t)$$

corresponde à soma dos pesos IPCW no tempo t.

O IBS é então definido como a integral do Brier Score ao longo do intervalo de acompanhamento:

$$IBS = \frac{1}{\tau}\int_0^{\tau} BS_{IPCW}(t)\,dt,$$

em que τ representa o tempo máximo de observação.

Por fim, a estimativa do EQM e do viés em relação à probabilidade dos não curados foi obtida a partir das seguintes expressões, respectivamente (PAL; ASELISEWINE, 2023):

$$\text{Viés}\left(\widehat{\pi}\left(\mathbf{x}\right)\right) = \frac{1}{N_1}\sum_{r=1}^{N_1}\left[\frac{1}{n}\sum_{i=1}^{n}\{\hat{\pi^{(r)}}\left(\mathbf{x}_i\right) - \pi^{(r)}\left(\mathbf{x}_i\right)\}\right],$$

$$\text{EQM}\left(\widehat{\pi}\left(\mathbf{x}\right)\right) = \frac{1}{N_1}\sum_{r=1}^{N_1}\left[\frac{1}{n}\sum_{i=1}^{n}\left\{\hat{\pi^{(r)}}\left(\mathbf{x}_i\right) - \pi^{(r)}\left(\mathbf{x}_i\right)\right\}^2\right].$$

CHAPTER

6

RESULTS

6.1 Aplicação a Dados Simulados

Nesta seção, avalia-se o desempenho do modelo proposto GG-KM em diferentes cenários de simulação, com o objetivo de investigar sua capacidade de adaptação sob diferentes estruturas do componente de cura. Para isso, foram considerados dois tamanhos de amostra ($n = 300$ e $n = 600$) e três métodos para geração de dados, representando diferentes relações entre covariáveis e fração de cura.

A Figura 5 apresenta a distribuição do IBS para cada kernel, obtida ao longo das replicações da validação cruzada. Como medida global de desempenho preditivo em análise de sobrevivência, menores valores de IBS indicam melhor ajuste.

De forma geral, observa-se que a escolha do kernel impacta diretamente a qualidade das estimativas, sobretudo em cenários mais complexos. No método 1, o desempenho entre os kernels é relativamente mais heterogêneo, com vantagem para kernels como cauchy, exponential e sigmoid, que apresentam menores valores centrais de IBS e menor variabilidade.

Nos métodos 2 e 3, por outro lado, o padrão é mais consistente: kernels não lineares, em especial rbf, cauchy, exponential e polynomial, tendem a apresentar melhor desempenho e maior estabilidade quando comparados ao kernel linear. Esse comportamento sugere que tais cenários envolvem estruturas mais complexas, cuja modelagem é favorecida por espaços de maior flexibilidade induzidos por esses kernels.

Dessa forma, os resultados mostram que o desempenho do GG-KM é fortemente dependente da especificação do kernel, sendo os kernels não lineares particularmente mais adequados em cenários com maior complexidade nos dados.

Como indicativo da capacidade do modelo proposto em capturar a fração de cura presente nos dados, foi realizada também uma comparação entre o estimador de Kaplan-Meier e a curva de

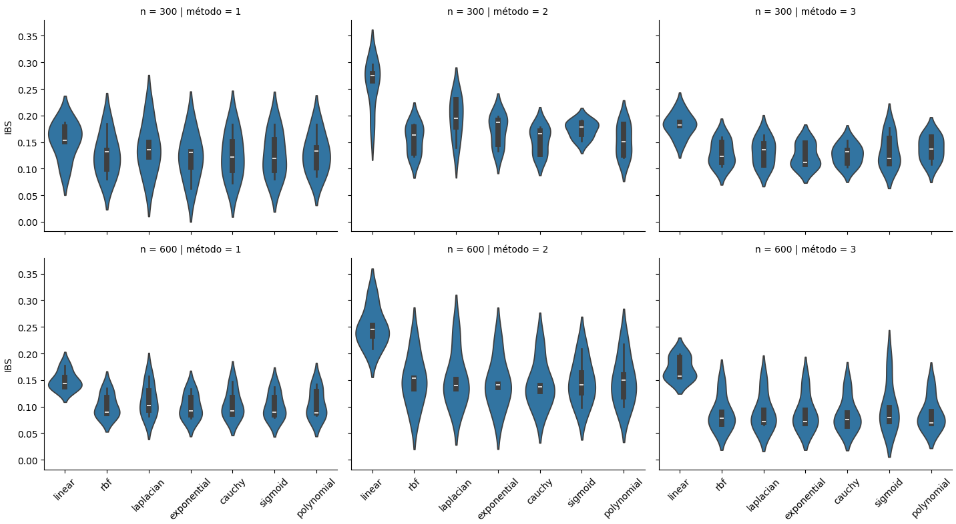

Figure 5 – Distribuição do IBS em relação a cada kernel, para diferentes cenários de simulação.

sobrevivência estimada pelo modelo GG-KM. Essa comparação pode ser observada na Figura 6. De modo geral, observa-se que o modelo GG-KM apresenta boa aderência ao comportamento do estimador de Kaplan-Meier, sugerindo que o componente de cura está sendo adequadamente modelado. Contudo, no cenário em que $n = 600$ para o método 3, nota-se um afastamento maior entre as curvas, indicando um ajuste inferior nesse caso específico.

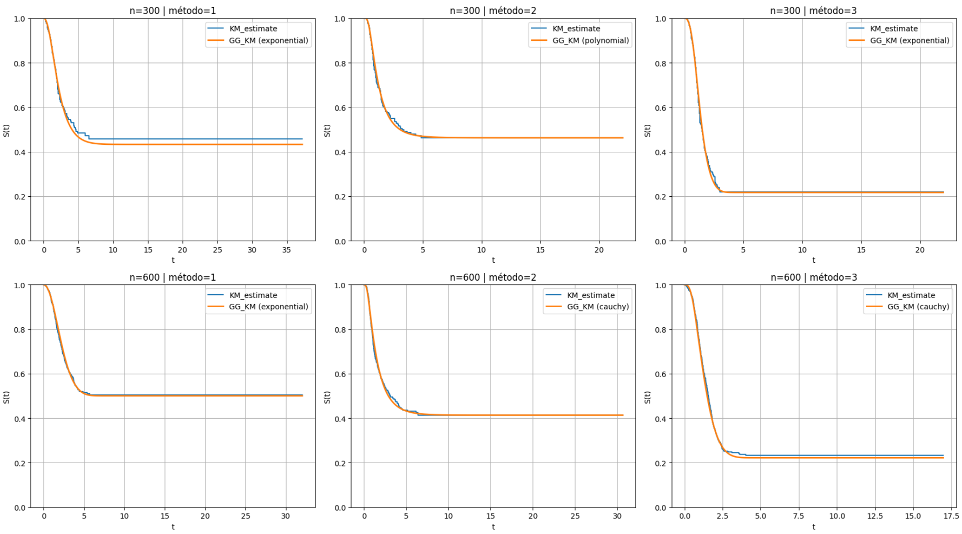

Figure 6 – Comparação da curva de sobrevivência estimada pelo GG-KM e KME.

A Tabela **??** complementa a análise anterior ao sintetizar, para cada cenário, o kernel com melhor desempenho segundo o critério de menor IBS médio. Os resultados reforçam a evidência de que kernels não lineares tendem a oferecer melhor capacidade de adaptação às diferentes estruturas simuladas.

Observa-se que o kernel cauchy foi selecionado nos cenários com $n = 300$ para os métodos 1 e 2, enquanto os kernels exponential, rbf, sigmoid e polynomial também se destacaram em diferentes configurações.

Além disso, nota-se que o aumento do tamanho amostral resultou, em geral, em redução do IBS médio e do viés absoluto, especialmente nos métodos 2 e 3, indicando ganhos em precisão e estabilidade na estimação da fração de cura. Em particular, o cenário com $n = 600$ e método 3 apresentou o melhor desempenho, com menor IBS e menor EQM.

Table 3 – Comparação entre as versões Poisson e Binomial Negativa do modelo GG-KM.

		Poisson				Binomial Negativa - com ϕ estimado			
n	Método	**K**	IBS	Viés $(\hat{\pi}(\mathbf{x}))$	EQM $(\hat{\pi}(\mathbf{x}))$	**K**	IBS	Viés $(\hat{\pi}(\mathbf{x}))$	EQM $(\hat{\pi}(\mathbf{x}))$
300	1	exp.	0.1221	-0.0505	0.0305	exp.	0.1220	-0.0443	0.0292
	2	poly.	0.1528	0.0709	0.0778	cauchy	0.1528	0.0539	0.0706
	3	exp.	0.1248	-0.0088	0.0211	exp.	0.1226	-0.0039	0.0197
600	1	exp.	0.1009	0.0100	0.0249	sigmoid	0.1008	0.0118	0.0300
	2	cauchy	0.1429	-0.0048	0.0260	exp.	0.1430	-0.0046	0.0297
	3	cauchy	0.0854	0.0052	0.0213	cauchy	0.0856	0.0052	0.0213

6.2 Aplicação a Dados Reais

O conjunto de dados reais escolhido para aplicação do modelo foi o Melanoma, amplamente utilizado em estudos de análise de sobrevivência. Esse conjunto contém informações de 205 pacientes diagnosticados com melanoma maligno, um tipo agressivo de câncer de pele, submetidos a cirurgia radical no Hospital Universitário de Odense, na Dinamarca, entre os anos de 1962 e 1977.

Os pacientes foram acompanhados até o final de 1977, período no qual 134 indivíduos permaneceram vivos, enquanto 71 evoluíram para óbito, sendo 57 mortes atribuídas diretamente ao melanoma e 14 decorrentes de outras causas. Este conjunto de dados, especificiamente, é utilizado em textos de modelos com fração de cura, pois há o conhecimento de que existe de fato uma fração de cura no conjunto de dados.

A base é composta por sete variáveis: tempo de acompanhamento em dias (time), status do paciente ao final do estudo (status), sexo (sex), idade no momento da cirurgia (age), ano da cirurgia (year), espessura do tumor em milímetros (thickness) e presença de ulceração (ulcer).

Para fins de modelagem, consideramos como evento de interesse o óbito por melanoma, tratando os demais casos, pacientes vivos ao final do estudo ou óbitos por outras causas, como observações censuradas. Portanto, ao final, tem-se aproximadamente 73% de observações censuradas.

Inicialmente, assim como foi feito com os dados simulados, foi realizada a comparação entre a curva de sobrevivência estimada pelo modelo proposto e o estimador de Kaplan-Meier, conforme apresentado na Figura 7. Como pode ser observado a partir da figura, há de fato indícios de que o modelo está conseguindo capturar adequadamente a fração de cura presente nos dados.

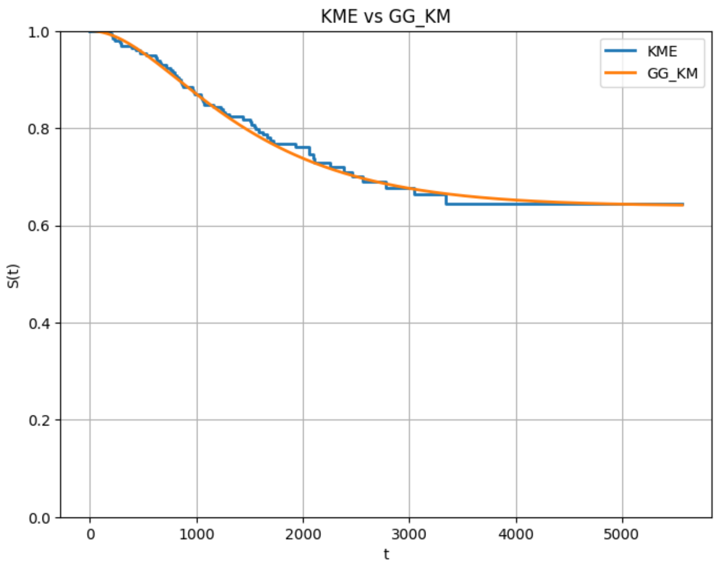

Figure 7 – Curva de Kaplan Meier para o conjunto de dados de melanoma, comparando o modelo GG-KM.

Na Figura 8, pode-se observar a variação do desempenho do modelo GG-KM para cada kernel, com base na distribuição do IBS. Nota-se que o kernel cauchy, embora tenha alcançado um dos menores valores de IBS, foi também o que apresentou maior variabilidade nessa métrica. Os kernels linear e gaussiano apresentaram desempenhos bastante semelhantes entre si. De forma análoga, os kernels laplaciano e exponencial também exibiram resultados próximos, destacando-se por apresentarem menor variabilidade, embora não tenham atingido os menores valores de IBS. Por fim, os kernels sigmoid e polinomial também apresentaram comportamento semelhante, tanto em termos de nível quanto de dispersão do erro.

A Tabela 4 resume esses resultados ao apresentar o IBS médio e seu respectivo desvio padrão para cada kernel. Observa-se que o kernel linear apresentou o menor IBS médio, indicando o melhor desempenho preditivo global nesse conjunto de dados, seguido de perto pelos kernels exponential e polynomial.

Embora o kernel cauchy tenha apresentado alguns bons resultados, seu IBS médio mais elevado e maior variabilidade indicam menor estabilidade no processo de estimação. Por outro lado, o kernel exponential se destaca por combinar desempenho preditivo próximo ao melhor observado com a menor dispersão entre todos os kernels avaliados, sugerindo maior robustez. De modo geral, os resultados indicam que, para o conjunto de dados Melanoma, estruturas mais simples ou moderadamente flexíveis parecem ser suficientes para capturar adequadamente a dinâmica da sobrevivência e a fração de cura presente nos dados.

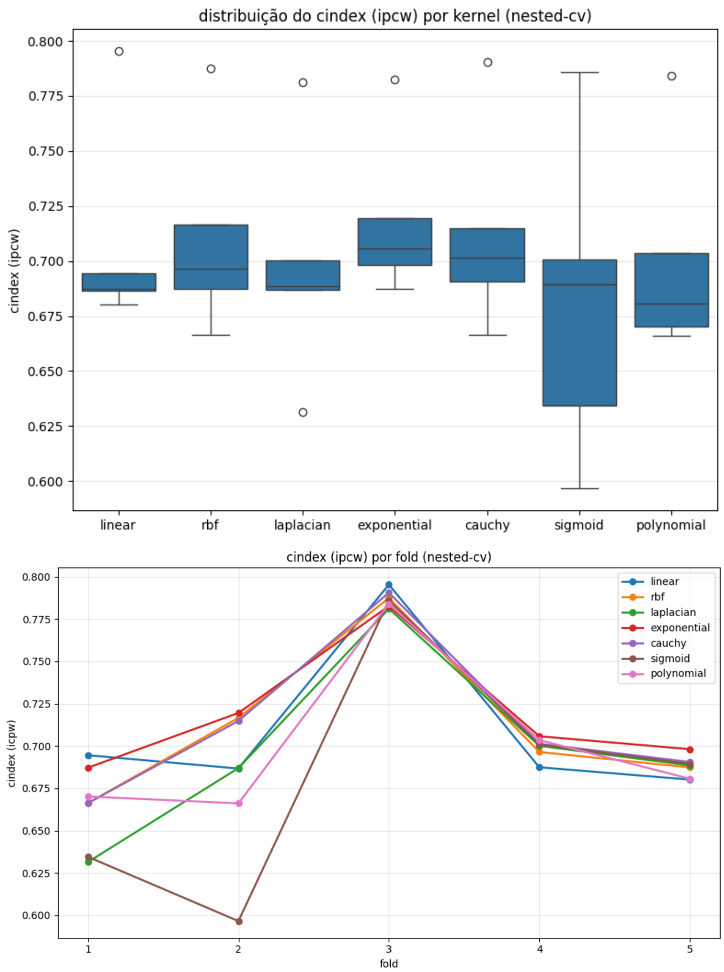

Figure 8 – Distribuição do IBS em relação a cada kernel.

Table 4 – Resultados do IBS médio, C-index médio, AUC médio e respectivos desvios padrão para cada kernel no conjunto de dados Melanoma, juntamente com o tempo de execução para cada distribuição de N.

N	Time (min)	Métrica	linear	exponential	cauchy	rbf	laplacian	polynomial	sigmoid
Poisson	45.90	IBS	0.16519	0.16203	0.16564	0.16363	0.16002	0.17562	0.16214
		DP IBS	0.02590	0.01918	0.02393	0.02289	0.02066	0.02836	0.01591
		C-index	0.68563	0.71525	0.69617	0.69732	0.71008	0.64992	0.71090
		DP C-index	0.09385	0.09151	0.09341	0.08980	0.09786	0.13254	0.08985
		AUC	0.81078	0.69441	0.73703	0.69553	0.70275	0.72194	0.68404
		DP AUC	0.05056	0.06351	0.05106	0.07890	0.03746	0.05533	0.06143
Binomial Negativa com ϕ estimado	68.46	IBS	0.16917	0.16203	0.16178	0.16131	0.15862	0.16293	0.16146
		DP IBS	0.02834	0.01916	0.02149	0.02114	0.02023	0.01676	0.01885
		C-index	0.70356	0.71488	0.70945	0.71009	0.71037	0.69918	0.70507
		DP C-index	0.08676	0.09212	0.09106	0.08934	0.09651	0.08920	0.09043
		AUC	0.76203	0.69459	0.71587	0.69384	0.69346	0.70382	0.69977
		DP AUC	0.03687	0.06344	0.03678	0.06031	0.02794	0.04479	0.06095
Bernoulli	291.26	IBS	0.16742	0.15948	0.16378	0.16364	0.15948	0.16393	0.16618
		DP IBS	0.03358	0.02212	0.02159	0.02224	0.02068	0.02494	0.02017
		C-index	0.68894	0.71595	0.70678	0.70468	0.70825	0.69844	0.69138
		DP C-index	0.09148	0.09836	0.08755	0.09047	0.09794	0.09558	0.09720
		AUC	0.83980	0.70202	0.70563	0.71164	0.70229	0.73775	0.66067
		DP AUC	0.05087	0.03374	0.05358	0.05173	0.05187	0.05418	0.08896
Binomial	86.60	IBS	0.16066	0.16184	0.16443	0.16231	0.15826	0.16660	0.15705
		DP IBS	0.01896	0.01983	0.02368	0.02268	0.02017	0.02120	0.02004
		C-index	0.67872	0.71194	0.69677	0.70904	0.70746	0.29430	0.72794
		DP C-index	0.09758	0.09374	0.09404	0.09510	0.10155	0.36978	0.09003
		AUC	0.54146	0.71091	0.71598	0.71328	0.66655	0.60220	0.70422
		DP AUC	0.04792	0.06211	0.05637	0.07016	0.09270	0.12809	0.06074

CHAPTER 7

CONCLUSION AND DISCUSSION

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APPENDIX

A

COMPLETE LOG-LOGLIKELIHOOD

A.1 Derivation of the Complete-Data Likelihood

This appendix derives the complete-data likelihood for the different distributions considered for the latent number of competing causes. The derivation is first presented in a general form and is then specialized to the Bernoulli, Binomial, Negative Binomial, and Poisson distributions.

A.1.1 Generic Complete-Data Likelihood

Recall that, for the ith individual, N_i denotes the latent number of competing causes and

$$T_i^* = \min\{Z_{i1}, \ldots, Z_{iN_i}\}, \tag{A.1}$$

where, conditional on $N_i = n_i$, the latent event times $Z_{i1}, \ldots, Z_{in_i}$ are assumed to be independent and identically distributed with proper survival function $S(t)$ and density $f(t)$.

Let C_i denote the censoring time, and define the observed time as

$$T_i = \min\{T_i^*, C_i\}. \tag{A.2}$$

The censoring indicator is denoted by δ_i, where $\delta_i = 1$ indicates that an event is observed at t_i, whereas $\delta_i = 0$ indicates that the observation is right censored at t_i.

First, conditional on $N_i = n_i$, the survival probability of the latent failure time T_i^* is

$$P(T_i^* > t_i \mid N_i = n_i) = P(Z_{i1} > t_i, \ldots, Z_{in_i} > t_i) = S(t_i)^{n_i}. \tag{A.3}$$

When $\delta_i = 1$, an event is observed at t_i, so that $T_i^* = t_i$. Since T_i^* is the minimum of the n_i latent event times, its conditional density is obtained by differentiating its survival function:

$$\begin{aligned} f(t_i \mid N_i = n_i) &= -\frac{d}{dt_i} S(t_i)^{n_i} \\ &= n_i f(t_i) S(t_i)^{n_i - 1}. \end{aligned} \tag{A.4}$$

When $\delta_i = 0$, the observation is right censored at t_i, which implies that $T_i^* > t_i$. Therefore, the corresponding conditional contribution is the survival probability given in Equation (A.3).

The complete data include the latent realization $N_i = n_i$. Thus, defining $\mathscr{D}_{\text{comp}} = \{(t_i, \delta_i, n_i) : i = 1, \ldots, n\}$, the contribution of the ith individual to the complete-data likelihood can be written as

$$L_{c,i} = P(N_i = n_i)\, P(T_i, \delta_i \mid N_i = n_i). \tag{A.5}$$

Combining the two possible values of δ_i gives

$$L_{c,i} = P(N_i = n_i) \left[n_i f(t_i) S(t_i)^{n_i - 1} \right]^{\delta_i} \left[S(t_i)^{n_i} \right]^{1-\delta_i}. \tag{A.6}$$

The first factor in Equation (A.6), $P(N_i = n_i)$, corresponds to the probability mass associated with the latent number of competing causes. The remaining factors describe the contribution of the observed event or censoring time conditional on this latent value.

Combining the powers of $S(t_i)$ gives

$$\begin{aligned} L_{c,i} &= P(N_i = n_i) n_i^{\delta_i} f(t_i)^{\delta_i} S(t_i)^{(n_i-1)\delta_i + n_i(1-\delta_i)} \\ &= P(N_i = n_i) n_i^{\delta_i} f(t_i)^{\delta_i} S(t_i)^{n_i - \delta_i}. \end{aligned} \tag{A.7}$$

Assuming independence across individuals, the complete-data likelihood is therefore

$$L_c\left(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}\right) = \prod_{i=1}^{n} P(N_i = n_i) n_i^{\delta_i} f(t_i)^{\delta_i} S(t_i)^{n_i - \delta_i}, \tag{A.8}$$

where $\boldsymbol{\eta}$ denotes the collection of model parameters.

Consequently, the complete-data log-likelihood is

$$\ell_c\left(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}\right) = \sum_{i=1}^{n} \left[\log P(N_i = n_i) + \delta_i \log n_i + \delta_i \log f(t_i) + (n_i - \delta_i) \log S(t_i) \right]. \tag{A.9}$$

The expression in Equation (A.9) provides the common structure for all distributions of the latent number of competing causes considered in this work. The distribution-specific complete-data log-likelihoods are obtained by substituting the corresponding probability mass function $P(N_i = n_i)$ into this expression.

A.1.2 Bernoulli Distribution

Suppose that

$$N_i \sim \text{Bernoulli}(\theta_i), \qquad 0 < \theta_i < 1, \tag{A.10}$$

so that

$$P(N_i = n_i) = \theta_i^{n_i} (1 - \theta_i)^{1 - n_i}, \qquad n_i \in \{0, 1\}. \tag{A.11}$$

Substituting Equation (A.11) into the generic complete-data likelihood in Equation (A.8) gives

$$L_c(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}) = \prod_{i=1}^{n} \theta_i^{n_i}(1-\theta_i)^{1-n_i} n_i^{\delta_i} f(t_i)^{\delta_i} S(t_i)^{n_i-\delta_i}. \tag{A.12}$$

Taking logarithms yields the complete-data log-likelihood

$$\ell_c(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}) = \sum_{i=1}^{n} \Big[n_i \log\theta_i + (1-n_i)\log(1-\theta_i) + \delta_i \log n_i + \delta_i \log f(t_i) + (n_i-\delta_i)\log S(t_i) \Big]. \tag{A.13}$$

Notice that when $\delta_i = 1$, an event is observed and therefore $n_i \geq 1$. In the Bernoulli case, this implies $n_i = 1$. Hence, the term $\delta_i \log n_i$ is well-defined for every observed event. When $\delta_i = 0$, this term vanishes and its value does not need to be evaluated.

A.1.3 Binomial Distribution

Now suppose that the latent number of competing causes follows a Binomial distribution, $N_i \sim \text{Binomial}(K, \theta_i^*)$, $0 < \theta_i^* < 1$, where K is a positive integer. Its probability mass function is

$$P(N_i = n_i) = \binom{K}{n_i} (\theta_i^*)^{n_i}(1-\theta_i^*)^{K-n_i}, \qquad n_i = 0, 1, \ldots, K. \tag{A.14}$$

Substituting Equation (A.14) into the generic complete-data likelihood in Equation (A.8) gives

$$L_c(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}) = \prod_{i=1}^{n} \binom{K}{n_i} (\theta_i^*)^{n_i}(1-\theta_i^*)^{K-n_i} n_i^{\delta_i} f(t_i)^{\delta_i} S(t_i)^{n_i-\delta_i}. \tag{A.15}$$

Taking logarithms yields the complete-data log-likelihood

$$\ell_c(\boldsymbol{\eta}; \mathscr{D}_{\text{comp}}) = \sum_{i=1}^{n} \left[\log\binom{K}{n_i} + n_i \log\theta_i^* + (K-n_i)\log(1-\theta_i^*) + \delta_i \log n_i + \delta_i \log f(t_i) + (n_i-\delta_i)\log S(t_i) \right]. \tag{A.16}$$

As in the Bernoulli case, when $\delta_i = 1$, an event is observed and therefore $n_i \geq 1$. Consequently, the term $\delta_i \log n_i$ is well-defined for all observations for which an event is observed. When $\delta_i = 0$, this term vanishes and does not need to be evaluated.

A.1.4 Negative Binomial Distribution

Consider the Negative Binomial parameterization adopted in this work, with $N_i \sim \text{NB}(\alpha, \theta_i)$, whose probability mass function is

$$P(N_i = n_i) = \frac{\Gamma(\alpha^{-1} + n_i)}{\Gamma(\alpha^{-1})\, n_i!} \left(\frac{\alpha\theta_i}{1+\alpha\theta_i} \right)^{n_i} (1+\alpha\theta_i)^{-1/\alpha}, \qquad n_i = 0, 1, 2, \ldots, \tag{A.17}$$

under the parameter restrictions required for this parameterization.

Substituting Equation (A.17) into the generic complete-data likelihood in Equation (A.8) gives

$$L_c\left(\boldsymbol{\eta};\mathscr{D}_{\text{comp}}\right)=\prod_{i=1}^{n}\left[\frac{\Gamma(\alpha^{-1}+n_i)}{\Gamma(\alpha^{-1})\,n_i!}\left(\frac{\alpha\theta_i}{1+\alpha\theta_i}\right)^{n_i}(1+\alpha\theta_i)^{-1/\alpha}\right.$$
$$\left.\times\, n_i^{\delta_i} f(t_i)^{\delta_i} S(t_i)^{n_i-\delta_i}\right]. \tag{A.18}$$

Taking logarithms and separating the terms gives the complete-data log-likelihood

$$\ell_c\left(\boldsymbol{\eta};\mathscr{D}_{\text{comp}}\right)=\sum_{i=1}^{n}\left[\log\Gamma(\alpha^{-1}+n_i)-\log\Gamma(\alpha^{-1})-\log(n_i!)\right.$$
$$+n_i\log\left(\frac{\alpha\theta_i}{1+\alpha\theta_i}\right)-\frac{1}{\alpha}\log(1+\alpha\theta_i)$$
$$\left.+\,\delta_i\log n_i+\delta_i\log f(t_i)+(n_i-\delta_i)\log S(t_i)\right]. \tag{A.19}$$

As in the Bernoulli and Binomial cases, when $\delta_i=1$, an event is observed and therefore $n_i\geq 1$. Hence, the term $\delta_i\log n_i$ is well-defined whenever it contributes to the log-likelihood. When $\delta_i=0$, this term vanishes and does not need to be evaluated.

A.1.5 Poisson Distribution

Finally, suppose that the latent number of competing causes follows a Poisson distribution, $N_i\sim\text{Poisson}(\theta_i)$, $\theta_i>0$. The corresponding probability mass function is

$$P(N_i=n_i)=\frac{\theta_i^{n_i}e^{-\theta_i}}{n_i!},\qquad n_i=0,1,2,\ldots. \tag{A.20}$$

Substituting Equation (A.20) into the generic complete-data likelihood in Equation (A.8) gives

$$L_c\left(\boldsymbol{\eta};\mathscr{D}_{\text{comp}}\right)=\prod_{i=1}^{n}\frac{\theta_i^{n_i}e^{-\theta_i}}{n_i!}n_i^{\delta_i}f(t_i)^{\delta_i}S(t_i)^{n_i-\delta_i}. \tag{A.21}$$

Taking logarithms yields the complete-data log-likelihood

$$\ell_c\left(\boldsymbol{\eta};\mathscr{D}_{\text{comp}}\right)=\sum_{i=1}^{n}\left[n_i\log\theta_i-\log(n_i!)-\theta_i+\delta_i\log n_i\right.$$
$$\left.+\,\delta_i\log f(t_i)+(n_i-\delta_i)\log S(t_i)\right]. \tag{A.22}$$

As in the previous cases, when $\delta_i=1$, an event is observed and therefore $n_i\geq 1$, so that the term $\delta_i\log n_i$ is well-defined. When $\delta_i=0$, this term vanishes and does not need to be evaluated.

Thus, the complete-data log-likelihoods for the four distributions considered in this work share the same contribution from the latent competing event times. They differ only through the probability mass function specified for the latent number of competing causes N_i.

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